

Chapter 19

Cycle processes

PREFACE

Chapter 17 already introduced a cycle process performed with a constant amount of gas in a cylinder–piston combination. This process enabled the transduction from thermal power into mechanical power like the process in a steam engine. This chapter will show that the concept of a cycle process is as domain independent as the concept it is performing the process, viz. the two–port storage element treated in the previous chapter. First a number of examples are given and the conclusions are generalized next.

OBJECTIVES

After studying these notes the reader is expected to be able to

- define a cycle process
- describe the energy transfer between two coupled coils as a cycle process with a two–port storage element
- describe the energy transfer between a coil that moves with respect to a magnetic field in terms of a cycle process with a two–port storage element
- recognize a cycle process with an arbitrary two–port storage element
- model, analyze and simulate a cycle process
- design a transducer that is based on a cycle process.

STUDY MATERIAL

1 Introduction

Cycle process

A *cycle process* is a process in which the *begin* state and *end* state are equal. Cycle processes play an important role in engineering. Each form of continuous *power transduction*—often, albeit not fully justified, called *energy transduction*—is based on a cycle process with a two–port storage element in principle. However, in many cases the energy storage can be neglected with respect to the transduction, such that only a power continuous transducer remains in the description. This transducer is, as shown earlier, either of the type *transformer*, or of the type *gyrator*. This chapter considers those cases where storage *cannot* be neglected. Next the conditions for neglecting the storage are considered. However, in practice a continuous power flow mostly cannot be realized by means of a component that is only described by a transformer or a gyrator, because a continuous flow often enters a storage element somewhere in the system, i.e. the flow is stored (integrated with respect to time, cf. chapter 4) into a state. Practical storage elements will always *saturate* at some point. This can be illustrated by means of an example in which the neglected storage is hard to identify, like in a gear transmission or a belt transmission. Here the transduction *seems* to take place *without storage*, but in practice the transduction will always take place via another domain—in this case the translation domain, viz. a small elastic deformation (translation) of the touching gear teeth or the belt. In case of a belt transmission this becomes most clear: a belt cannot be infinitely stiff in practice. The part of the belt that is being pulled on, will be

extended while it returns the stored energy during after rounding the driven wheel by contraction, not considering irreversible effects like slip. In case of the gearbox the teeth serve as elastic storage elements in the period in which they are in contact to each other. In other words: by putting storage elements in a specific configuration such that they are part of a cycle process, it becomes possible to create a net power transduction. Obviously this does not mean that the *elements* transformer (TF) and gyrator (GY) would not be useful concepts, but it is always good to keep in mind that realization of a transducer as a *component* requires a cycle process at some point. Both during engineering design and diagnosis this may result in useful information.

Another example is the electric transformer that consists in practice of two coupled coils. It can only transduce power if the port signals describe a *cycle* (in time), in other words: if periodic signals are concerned. By the way, in this case the process is called transduction too, although both ports of this transducer belong to the same type of domain (electric). By the galvanic separation of the circuits at both sides of the transformer —i.e. the circuits are not connected by means of conductors— it is possible to give these circuits different references. During the introduction of the systematic method for constructing a bond graph it was noted that each domain needs a choice of a reference. If the reference were used as an unambiguous criterion for identifying a domain, the two circuits would belong to different domains. A similar remark can be made for translations in orthogonal directions. In this case a reference has to be chosen for each individual direction (degree of freedom). The above discussion of transduction also supports this point of view. It is at least useful to interpret the concept *domain* in this way during application of the systematic method.

The previous chapter (formula (18.52)) demonstrated that two coupled coils can be described in most cases by linear constitutive relations. Therefore, this example is elaborated first. Next the examples of a coil in a magnetic field (like in a loudspeaker), an electret microphone, a piezo–electric transducer and finally a coil with a moving core are considered. Cycle processes like those present in the electric energy generation and in heating and cooling systems are discussed shortly too. The description is generalized to the domain independent case and in principle only C–type (*q*–type state) storing ports are used. The analysis of I–type (*p*–type state) storing ports is fully dualog and therefore not separately elaborated, with the exception of the coupled coils that can be treated as a magnetic two–port C, but are treated as an electric two–port I in order to be in line with the common terminology for this case. In this particular example the dualization causes no problem.

2 The electric transformer: two coupled coils

A transformer is described by the following constitutive equations (18.52):

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} \quad (19.1a)$$

or

$$\begin{aligned} u_1 &= L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \\ u_2 &= M \frac{di_1}{dt} + L_2 \frac{di_2}{dt} \end{aligned} \quad (19.1b)$$

If $M = 0$ two separate one-port coils remain, whereas there is coupling if $M \neq 0$. Analog, or rather dualog, to the capacitor with moving plate a two-port I-type storage element can be introduced. The explanation given earlier of the fact that an electric transformer consisting of two coupled coils does not transform DC voltage and current is elaborated in terms of a cycle process with this two-port storage element. First periodic signals that allow a simple analysis are considered, viz. block shaped voltages at the ports of the coils (figure 19.1). The indices 1 and 2 are assigned to these ports for identification. Each period is divided in four equal parts. In the first part, section AB, the port voltages are $u_1 = +10$ V, $u_2 = 0$ V. In the second part BC the port voltages are $u_1 = 0$ V and $u_2 = +10$ V, in the third part CD $u_1 = -10$ V and $u_2 = 0$ V and finally in the last part DA $u_1 = 0$ V and $u_2 = -10$ V, after which this cycle repeats itself. Each part takes 0.1 second, such that the total period is 0.4 seconds. The coils are without current in point A.

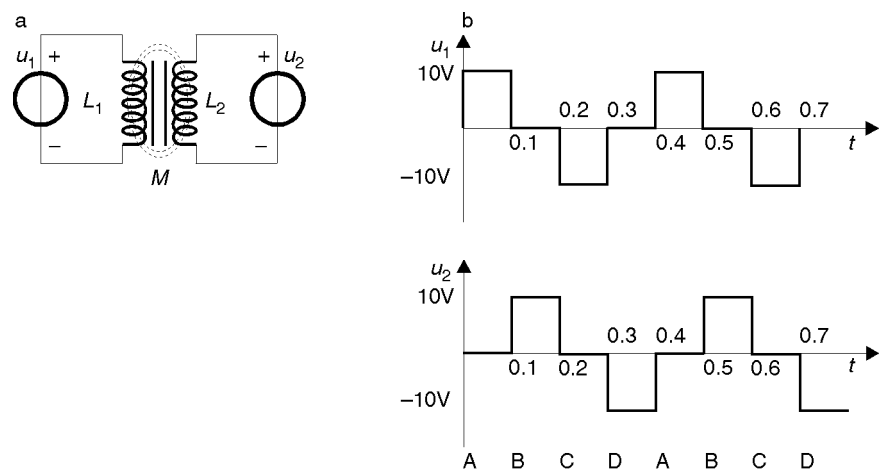


FIGURE 19.1 Transformer
a circuit
b signals

ASSIGNMENT 19.1

Give a causal bond graph of this situation. First use a two-port I-type storage element and convert this into a two-port C-type storage element by making the coupling between the electric and the magnetic domain explicit. Sketch the flux linkages of the coils as a function of time.

In order to be able compute the currents through the coils, the constitutive relations need to be known. For reasons of (computational) simplicity the following—not quite realistic—values in the constitutive relation (19.1a) are chosen: $L_1 = 2$ H, $L_2 = 1$ H (self inductions) and $M = 1$ H (mutual induction). In this linear case the results can always be adapted later to realistic parameter values by means of scaling. The shape of the signal will not change, due to the linearity of the system.

ASSIGNMENT 19.2

Compute and sketch the currents through the coils as a function of time.

In order to make clear that a cycle process is involved, it is instructive to sketch the bundles of characteristics (cf. figure 18.2) and to indicate the four parts of a period.

ASSIGNMENT 19.3

Sketch the bundles of characteristics of both ports and indicate the parts AB, BC, CD and DA. Use the result to derive which port absorbs power during one cycle and which port delivers power. Compute the amount of energy graphically.

It also provides insight to plot the behavior of the power and of the stored energy as a function of time.

ASSIGNMENT 19.4

Sketch the power of both ports and the stored amount of energy as a function of time.

Purely block-shaped voltages are commonly not used in practice but sine-shaped signals are common. From the above sketch it can be concluded that when a sine-shaped or a cosine-shaped voltage is put at the ports respectively, the derivation of completely analogous, but that the signals and the cycles in the port characteristics will be more smooth, because the signals do not contain discontinuities anymore.

ASSIGNMENT 19.5

Perform the first two experiments in chapter 24 corresponding to chapter 19.

On the basis of the example of the two coupled coils the importance of the distinction between energy and co-energy is considered again and the coupling factor is expressed in the constitutive parameters.

2.2 THE ENERGY AND CO-ENERGY OF TWO COUPLED COILS

The energy of two coupled coils can be found by substitution of the constitutive relation (19.1a) into the standard form of the total differential of the energy of this two-port storage element, being:

$$dE = i_1 d\lambda_1 + i_2 d\lambda_2 \quad (19.2)$$

where the constitutive matrix has to be inverted (cf. solution to assignment 18.15) and by integrating the result next:

$$E(\lambda_1, \lambda_2) = \frac{1}{D} \left(\frac{1}{2} L_2 \lambda_1^2 - M \lambda_1 \lambda_2 + \frac{1}{2} L_1 \lambda_2^2 \right) \quad (19.3)$$

where $D = L_1 L_2 - M^2$, the determinant of the constitutive matrix.

For the coupling factor k (18.58) holds in this case:

$$k = \sqrt{\frac{M^2}{L_1 L_2}} \quad (19.4)$$

This storage element is intrinsically stable if the constitutive relations satisfy the usual conditions (cf. section 18.5), viz. $L_1 > 0$, $L_2 > 0$ and $0 < k < 1$ or $D > 0$. If the currents are to be imposed, for example for the design of a cycle process, in other words when the currents are chosen as independent variables, this means that a Legendre transform should be performed. In turn, this means in a bond graph representation that both storing ports obtain a differential causality. The Legendre transform of the energy with respect to λ_1 and λ_2 results in a co-energy $E^*(i_1, i_2)$. If the differential form (19.2) is taken as a starting point the total differential of the co-energy becomes

$$dE^*(i_1, i_2) = \lambda_1 di_1 + \lambda_2 di_2 \quad (19.5)$$

Combination with (18.52) and integration results in:

$$E^*(i_1, i_2) = \frac{1}{2} L_1 i_1^2 + M i_1 i_2 + \frac{1}{2} L_2 i_2^2 \quad (19.6)$$

In the linear case this co-energy is obviously equal in value again to the energy. For non-linear constitutive relations, for example in case of a realistic representation of saturation, this does not hold.

In order to be able to describe, for instance, a coil in a magnetic field and in order to show which conditions are needed to transform the storing model of two coupled coils into a TF element, it is useful to elaborate a number of aspects of the already derived model of two coupled coils. First the two-port storage element is decomposed into a conceptual structure that only contains basic elements. Next the concepts mutual energy and interaction energy are discussed in the light of the already known coupling factor (18.58), in order to reveal which terms in the expression of the energy need a separate treatment in the examples still to be discussed and what happens if the coupling factor becomes 1.

2.3 DECOMPOSITION OF A LINEAR TWO-PORT STORAGE ELEMENT

This chapter intends to give more insight into two-port storage elements. A linear two-port storage element is equivalent, as for port behavior, to the following bond graph.

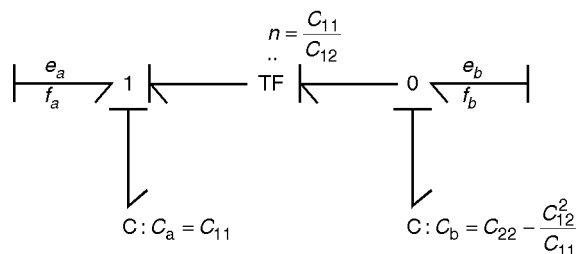


FIGURE 19.2a Bond graph of which the port behavior is equivalent to that of a linear two-port storage element

The two external ports are given the labels a and b . The values of the C-elements can also be expressed in terms of the coupling factor k (18.58):

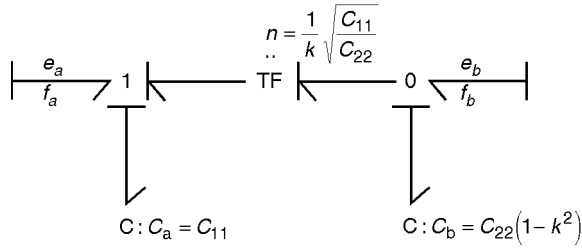


FIGURE 19.2.b Bond graph of figure 19.1 with values of C-elements expressed in terms of the coupling factor k

ASSIGNMENT 19.6

Proof that the two-port structure in figure 19.2 is equivalent, as for port behavior, to a two-port storage element by writing the corresponding equations in matrix form like: $se = Cf$, where s is the Laplace operator.

Finding these kinds of *decompositions* is elaborated in chapter 21. Figure 19.3 shows the same result for the case of the two coupled coils.

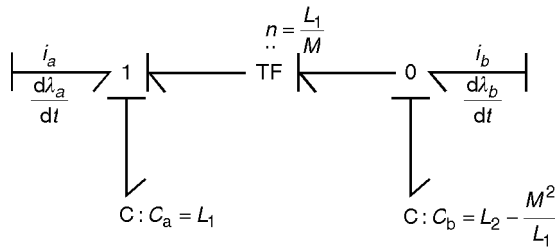


FIGURE 19.3 Decomposition of a multiport C-type storage element representing two coupled coils

Finally note that, if the constitutive parameters are defined as the matrix elements of a partially inverted constitutive matrix (i.e. port a with integral causality and port b with differential causality), these matrix elements represent the constitutive parameters of the elements in the decomposition almost directly. Starting from the form with differential causality

$$\begin{aligned} f_a &= sC_{11}e_a + sC_{12}e_b \\ f_b &= sC_{12}e_a + sC_{22}e_b \end{aligned} \quad (19.7)$$

causal inversion of port a results in:

$$\begin{aligned} e_a &= \frac{1}{sC_{11}} f_a - \frac{C_{12}}{C_{11}} e_b = \frac{1}{sC_a} f_a - \frac{1}{n} e_b \\ f_b &= \frac{C_{12}}{C_{11}} f_a - s \frac{C_{12}^2}{C_{11}} e_b + sC_{22}e_b = \frac{C_{12}}{C_{11}} f_a + sC_{22} \left(1 - \frac{C_{12}^2}{C_{11}C_{22}} \right) e_b = \frac{1}{n} f_a + sC_b e_b \end{aligned} \quad (19.8)$$

The first term at the diagonal equals the inverse of the left storage element parameter and the second equals the right storage element parameter. The modulus of the off-diagonal terms equals the inverse of the transformation ratio. This result will be used again to quickly find a decomposition in cases where a linear or linearized constitutive relation with a mixed causality is to be derived, e.g. in the examples of an electret microphone and a coil with moving core to be treated later in this chapter.

2.4 MUTUAL ENERGY AND COUPLING FACTOR

Although the concept of mutual energy has been introduced generically for a two-port storage element, its treatment has been postponed until the treatment of the two coupled coils, because this is an illustrative example of this concept. It is shown how the coupling factor can be expressed in terms of the mutual energy, because this expression often serves as its definition.

If the constitutive relations of a two-port C are linear or linearized into the form

$$\begin{aligned} e_1 &= a_{11}q_1 + a_{12}q_2 \\ e_2 &= a_{12}q_1 + a_{22}q_2 \end{aligned} \quad (19.9)$$

its energy can be written

$$E(q_1, q_2) = E(\mathbf{q}) = \frac{\mathbf{q}^T \mathbf{A} \mathbf{q}}{2} = \frac{a_{11}q_1^2}{2} + a_{12}q_1q_2 + \frac{a_{22}q_2^2}{2} \quad (19.10)$$

If the coupling factor k (where $k^2 = a_{12}^2/a_{11}a_{22}$) equals zero, in other words if $a_{12} = 0$, the storage element degenerates into two separate one-ports. In that case—and only in that case—the energy can be assigned to the corresponding port. Suppose that port 1 is mechanical and port 2 electric, like in the case of the flat-plate-capacitor. Then the term $a_{11}q_1^2/2$ represents the mechanical energy and the term $a_{22}q_2^2/2$ the electric energy. As soon as a_{12} does not equal zero the separate terms cannot be interpreted as the energy belonging to a specific domain. *Nevertheless* this is often encountered: half of the term $a_{12}q_1q_2$, i.e. $a_{12}q_1q_2/2$, is then called the *mutual energy* and the two remaining terms are coupled to the domain to which they can be assigned in the case that $a_{12} = 0$. The term $a_{ii}q_i^2/2$ is also called the *self energy* of port i . The following relation can be derived for k :

$$k^2 = \frac{a_{12}^2}{a_{11}a_{22}} = \frac{\frac{a_{12}^2q_1^2q_2^2}{4}}{\left(\frac{a_{11}q_1^2}{2}\right)\left(\frac{a_{22}q_2^2}{2}\right)} = \frac{E_{\text{mutual}}^2}{E_{\text{electric}}E_{\text{mechanical}}} \quad (19.11)$$

alternatively:

$$k = \left(\frac{E_{\text{mutual}}^2}{E_{\text{electric}}E_{\text{mechanical}}} \right)^{0.5} = \frac{E_{\text{mutual}}}{\sqrt{E_{\text{electric}}E_{\text{mechanical}}}} \quad (19.12)$$

Mutual energy

Self energy

This is mentioned herein, because this is also considered to be the definition of the coupling factor. However, note that the terms E_{mutual} at the one hand and E_{electric} and $E_{\text{mechanical}}$ at the other hand are somewhat contradictory: E_{electric} and $E_{\text{mechanical}}$ can only be characterized like that when E_{mutual} equals zero. By the way, the mutual energy can be interpreted as follows: if both storing ports are decoupled, the total energy does not contain this term. After the ‘coupling process’ of the ports the total energy has increased with twice this term. This means an ‘addition’ of subsystems (storage elements) that is not interaction-free. The mutual energy should be released again when the subsystems are separated again. This is why this energy, that is twice the mutual energy, is also called *interaction energy*. In case this separation is indeed a spatial ‘pulling apart’, this energy should be transferred in the form of mechanical energy. If the coils in the example of the two coupled coils can move with respect to each other, this allows the following analysis. The amount of work A that needs to be performed in order to bring the coils from a situation where the mutual induction $M = 0$ into a situation where $M \neq 0$ has to be equal to the interaction energy. From (19.3) and (19.4) follows for this amount of work:

$$\begin{aligned} A &= -\frac{M}{D} \lambda_1 \lambda_2 = -\frac{M^2}{L_1 L_2 - M^2} \frac{\lambda_1 \lambda_2}{M} \\ &= -\frac{k^2}{1-k^2} \frac{\lambda_1 \lambda_2}{M} = -\frac{k}{1-k^2} \frac{\lambda_1 \lambda_2}{\sqrt{L_1 L_2}} \end{aligned} \quad (19.13)$$

Note that this amount of work is negative when λ_1 and λ_2 have equal signs (this means that the coils attract each other) and positive when λ_1 and λ_2 have different signs (work needs to be performed to bring the mutually repulsive coils together). When the coil currents i_1 and i_2 are kept constant, an analysis based on co-energy leads to the same result: the work equals in that case $-Mi_1 i_2$, where the minus sign is caused by the Legendre transform. This can also be interpreted as follows: the only parameter that depends on x is $M = M(x)$, such that

$$\begin{aligned} F(i_1, i_2, x) &= -\frac{\partial E^*(i_1, i_2, x)}{\partial x} \\ &= -\frac{\partial}{\partial x} \left(\frac{1}{2} L_1 i_1^2 + M(x) i_1 i_2 + \frac{1}{2} L_2 i_2^2 \right) = -i_1 i_2 \frac{dM(x)}{dx} \end{aligned} \quad (19.14)$$

The total differential of the work is:

$$dA|_{i_1, i_2} = F|_{i_1, i_2} dx = -i_1 i_2 \frac{dM(x)}{dx} dx = d(-M(x) i_1 i_2) \quad (19.15)$$

After having considered the concept of the coupling factor from different angles, it will now be analyzed what happens when the coupling factor approaches 1, i.e. full coupling of the storing ports, in this case the coils.

2.5 FULL COUPLING

If both coils are wound on top of each other such that they may be considered to fully enclose the same flux Φ , the flux linkages of both coils are no independent states anymore. Suppose that the first coil has n_1 windings and the second n_2 :

$$\lambda_1 = n_1 \Phi = L_1 i_1 + M i_2 \quad (19.16)$$

$$\lambda_2 = n_2 \Phi = M i_1 + L_2 i_2 \quad (19.17)$$

This results in

$$\frac{L_1}{n_1} i_1 + \frac{M}{n_1} i_2 = \frac{M}{n_2} i_1 + \frac{L_2}{n_2} i_2 \quad \text{for all values of } (i_1, i_2) \quad (19.18)$$

so

$$\frac{L_1}{n_1} = \frac{M}{n_2} \quad \text{and} \quad \frac{M}{n_1} = \frac{L_2}{n_2} \quad \text{or} \quad \frac{L_1}{M} = \frac{n_1}{n_2} \quad \text{and} \quad \frac{L_2}{M} = \frac{n_2}{n_1}, \quad \text{such that } M^2 = L_1 L_2.$$

In this case the coupling factor (19.4) is $k = \sqrt{\frac{M^2}{L_1 L_2}} = 1$. This is a singular case

that expresses the dependence of the states λ_1 and λ_2 . In this case this means that only one independent state Φ is left, which can be represented by a one-port C. This is also shown in the decomposition derived earlier, where the

transformation ratio equals $\frac{L_1}{M} = \frac{n_1}{n_2}$ and where the storage element at the right-hand side disappears (figure 19.4).

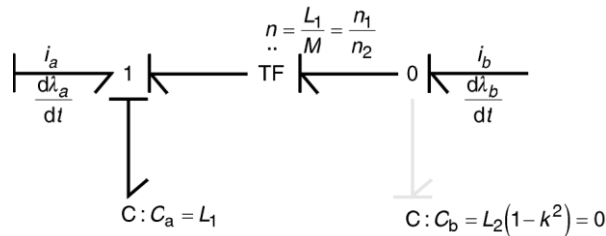
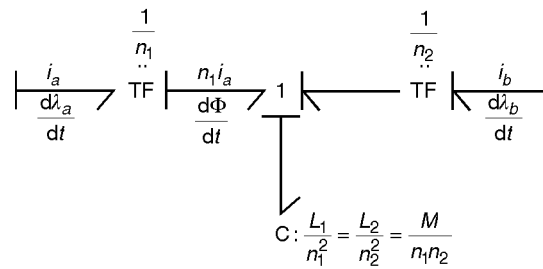


FIGURE 19.4 The storage element at the right-hand side disappears if $k = 1$.

Note that the causality of the storage element at the right-hand side in figure (19.4) should become differential in order to enable the constitutive parameter to become zero. (In case of integral causality the parameter C is in the denominator of the constitutive relation.) This also demonstrates the dependence of the flows of the both external ports. Either by manipulating the equations or by manipulating the bond graph it can be demonstrated that the model becomes a one-port C with capacitance $L_1/n_1^2 = L_2/n_2^2 = M/n_1 n_2$ to a 1-junction that represents $d\Phi/dt$ and two transformers with transformation ratios $1/n_1$ and $1/n_2$ (figure 19.5a).

FIGURE 19.5a Decomposition of two coupled coils in case $k = 1$

It is demonstrated how figure 19.4 can be transformed into figure 19.5a. First figure 19.4 is drawn again (19.5b):

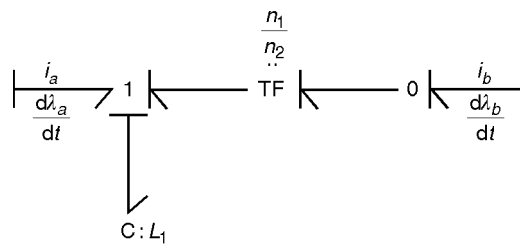


FIGURE 19.5b Figure 19.4 simplified

The first step is the reversal of the orientation of the bonds at the TF. This changes the transformation ratio from n_1/n_2 into $-n_2/n_1$.

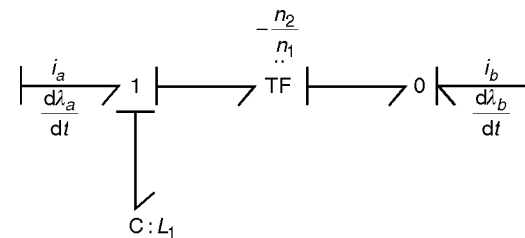


FIGURE 19.5c Change in orientation of a bond

Next the TF is split:

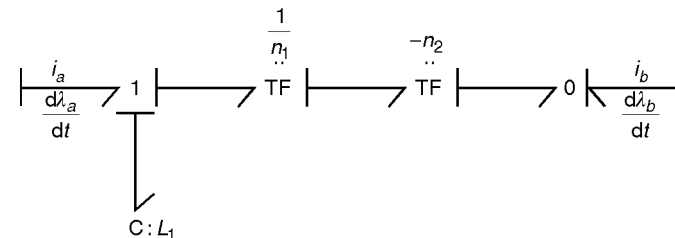


FIGURE 19.5d Separation of the TF

In the following step the C-element is put in between the two transformers:

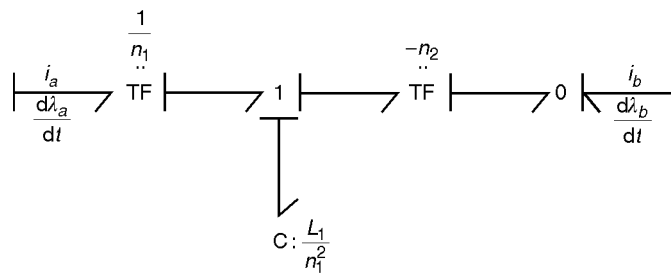
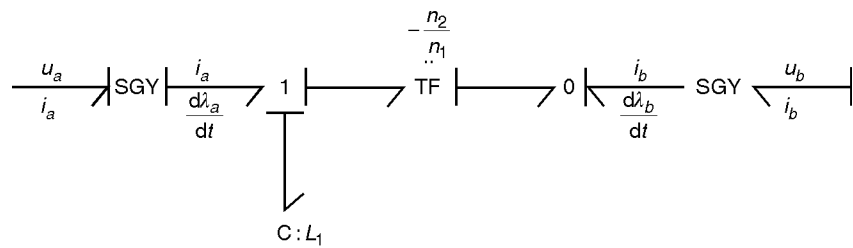


FIGURE 19.5c Moving the C–element between the transformers

The final step changes the orientations of the bonds such that the positive power orientation is towards the C–element. This gives the final result in figure 19.5a.

In order to represent the coupling of the magnetic ports in the electric domain, symplectic gyrators are added to the bond graph in figure 19.5c (figure 19.6a).

FIGURE 19.6a Figure 19.5c with symplectic gyrators
(transition from the magnetic to the electric domain)

Next these SGY's can be eliminated by partial dualization, such that the complete bond graph is expressed in terms of electric variables (figure 19.6b).

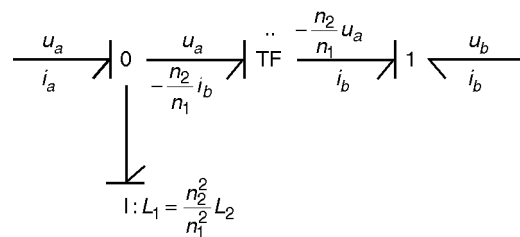


FIGURE 19.6b Dualization of figure 19.6a

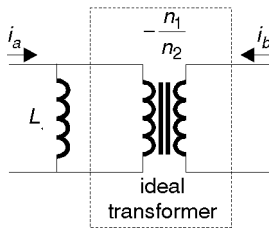


FIGURE 19.7
Electric diagram corresponding to
figure 19.6b

This figure corresponds to an electric equivalent circuit diagram consisting of an ideal coil with self induction L_1 , that is connected in parallel with the primary port of an ideal transformer with transformation ratio $-n_1/n_2$ (figure 19.7). The figures 19.6b and 19.7 show that if the storing behavior of this fully coupled two-port storage element is neglected, i.e. that L_1 disappears, a transforming transducer is left with transformation ratio $-n_1/n_2$, where the current i_b through the secondary coil has a sign that is opposite to the sign of the current i_a through the primary coil with equal orientation of the windings:

$$i_b = -\left(\frac{n_1}{n_2}\right)i_a$$

In accordance with the earlier announcement the relation between the two-port storage element model of two coupled coils and the model consisting of the basic TF-element has been demonstrated: two coupled coils can be described by a TF in case the coupling can be considered to be full and the storage can be neglected. The latter is often not the case, as demonstrated by the fact that a TF allows DC transformation, while the addition of the I-type storage element with a 0-junction demonstrates that direct current (DC) is short-circuited via this coil. After all if the DC case is considered the voltage over the coil is $u = L_1(di_a/dt) = 0$.

3 Coil in a magnetic field

An example that is closely related to the former example of the two coupled coils is the case in which one of the coils is replaced by a magnetic field that can be generated in an arbitrary manner (figure 19.8). However, it is assumed that the influence of the coil at this field can be neglected. This assumption suggests that one does not need to describe the storage of the external field as such analog to the model of a battery in chapter 3 and that its behavior can be replaced by a source accordingly. Chapter 3 also demonstrated that the transition of a storage element in a source can be interpreted as the constitutive parameter of the storage element approaching an infinite value. As a *two-port storage element* is considered here, the analysis is somewhat more complex, at least as for the derivation.

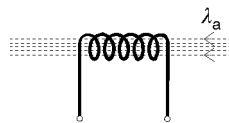


FIGURE 19.8
Coil in a magnetic field of arbitrary
origin

Applied flux

The field may be generated by a permanent magnet for instance. For the time being it is assumed however that the external field is generated by a coil of which neither the self-induction L_2 (that should be large compared to L_1 , such that $L_2 i_2 \gg M i_1$) nor the mutual induction M with the first coil (with self induction L_1) are known. A description of the effect of such a field often used in practice is by means of the coupled flux that is measured in the coil when the coil is without current. This coupled flux is called *applied flux* λ_a . Note that this is an *operational* definition of the applied flux that is *not* an *actual* variable of the system: this variable is only hypothetically measurable, viz. when it would be possible to make the current in the coil zero at that moment.

ASSIGNMENT 19.7

What is the applied flux in the case that the field is applied by a second coil with self induction L_2 and mutual induction M ?

The solution of assignment 19.7 shows that the constitutive relation of the first coil can be rewritten

$$\lambda_1 = L_1 i_1 + \lambda_a \quad (19.19)$$

Next the effect of the assumptions at the model of the two coupled coils is studied and in particular at their decomposition shown in figure 19.3. This figure is repeated in 19.9a.

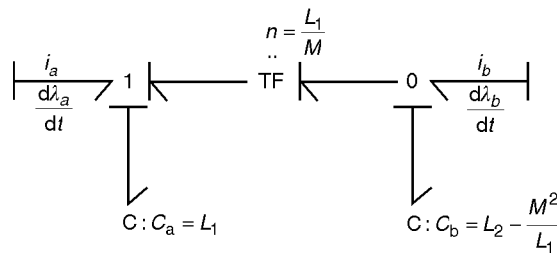


FIGURE 19.9a Figure 19.3

An attempt is made first to make a separation in the decomposition between at the one hand as many as possible elements characterized by the known parameter L_1 and at the other hand elements that are characterized by the unknown parameters M and L_2 (figure 19.9b). (The external ports are given the labels 1 and 2 in stead of a and b , in order to prevent confusion with the subscript a of *applied* field.)

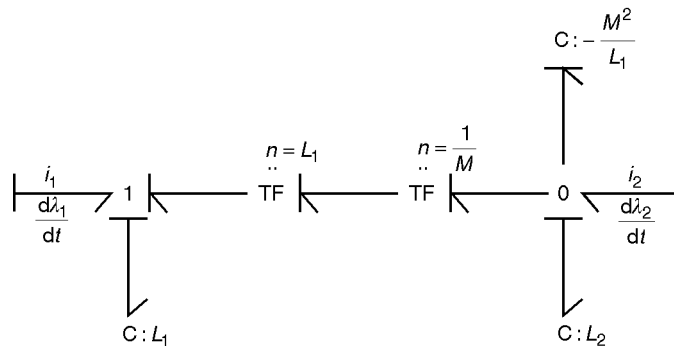


FIGURE 19.9b Separation of the TF and C_b

Next the elements characterized by L_1 are moved left by transformation and the elements characterized by M and L_2 are moved right in order to achieve a separation (figure 19.9c).

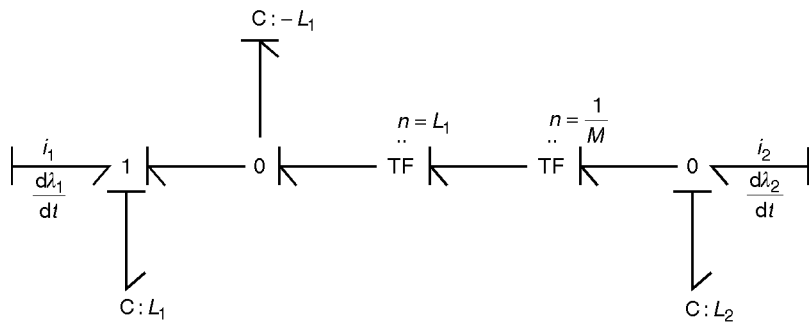


FIGURE 19.9c Transformation of the negative C over both TF's

Finally the bond graph in figure 19.9d results.

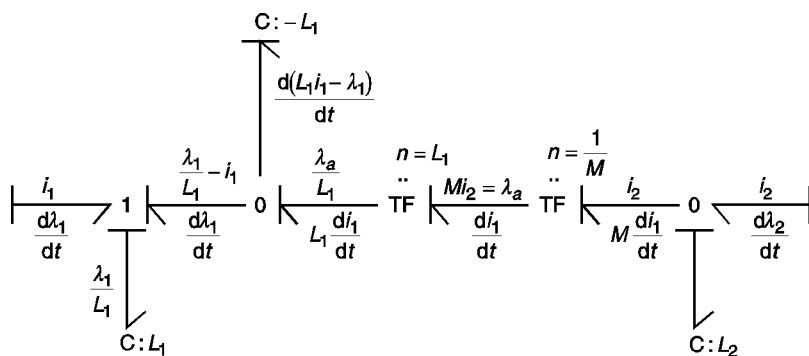


FIGURE 19.9d The final result with all relevant bond variables

Figure 19.9d clearly shows that the bond that connects the ‘known’ part with the ‘unknown’ part is characterized by the conjugate variable pair $(\lambda_a, di_1/dt)$, where di_1/dt is the flow variable, such that the variable i_1 at the magnetic port of the coil $(i_1, d\lambda_1/dt)$ plays the role of an effort, whereas it has the nature of a q -type state (integrated flow) at this port inside the conceptual structure. This is precisely why the analysis of a coil in an external field that is not complex by itself often causes so much confusion. After the following analysis it will be demonstrated that this confusion can be prevented by not characterizing the external field by the variable λ_a . To that end figure 19.9d shows the effect of the assumption of a large value of L_2 with respect to L_1 . If L_2 is allowed to approach infinity, the C-type storage element at the right-hand side an Se-source (figure 19.10).

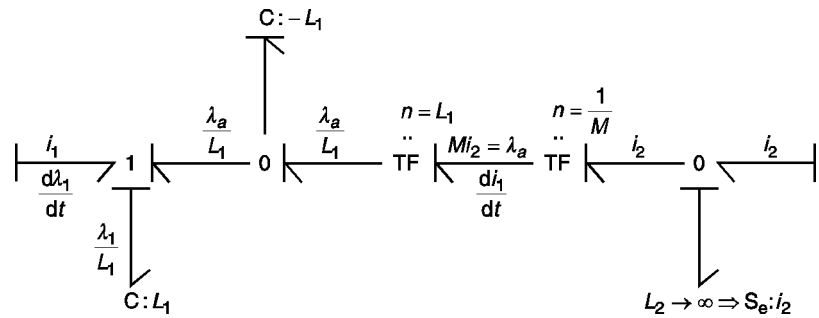


FIGURE 19.10a If $L_2 \rightarrow \infty$ the C-type storage element at the right-hand side becomes an Se-source

As a result, the external port at the right-hand side has no influence anymore, such that it can be omitted and the source can be combined with the TF at the right-hand side. The resulting bond graph is shown in figure 19.10b.

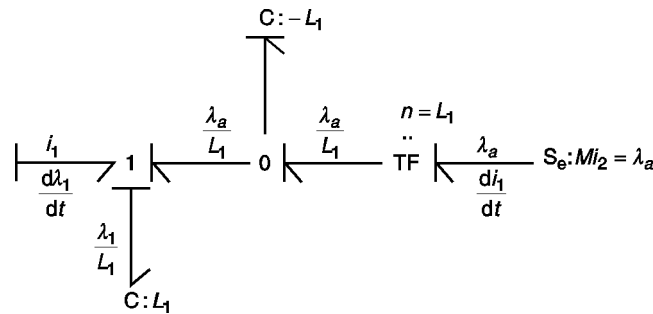


FIGURE 19.10b Simplification of the bond graph in figure 19.10a

The resulting effort source supplies λ_a . However, the Se-source can be taken a step further and combined with the left TF too (figure 19.10c).

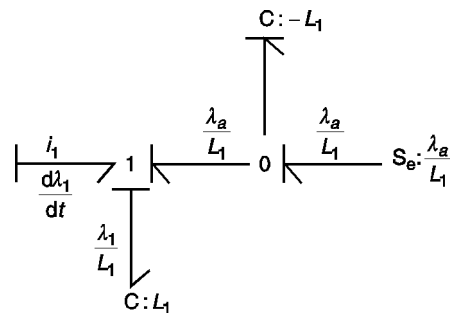


FIGURE 19.10c Elimination of the second TF

That way it becomes even more transparent that, due to the influence of this source that supplies a voltage λ_a/L_1 , the storage element with differential causality and $-L_1$ as parameter also does not influence the rest of the model anymore. It can thus be omitted, such that the simple magnetic model in figure 19.10d remains.

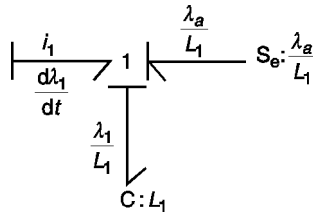


FIGURE 19.10d
Omitting the storage element with
differential causality

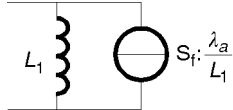


FIGURE 19.10f
Equivalent electric circuit diagram

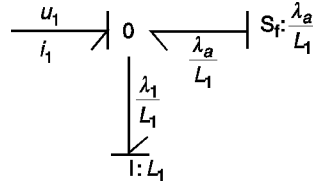


FIGURE 19.10e
Dualization

This shows that the influence of the external field is in this case completely described by a parallel current source that supplies an additional current λ_a/L_1 . This interpretation shows clearly that it is not possible in this case to work both with the assumption for the external field that leads to the use of the variable λ_a and with a two-port model at the same time. The assumption means that the influence of the second port disappears due to the fact that a storage element becomes a source, letting only a one-port remain. If the system approach introduced in this text is not (sufficiently) known, the bond in figure 19.9d characterized by the conjugate variable pair $(\lambda_a, di_1/dt)$ can easily be mistaken, for the second port of a storage element. In that case the following, rather awkward, ‘energy relation’ is found [Fluitman, 1980]:

$$d\tilde{E}^* = i_1 d\lambda_1 + \lambda_a di_1 \left(= i_1 d\lambda_1 + \frac{\lambda_a}{L_1} dL_1 i_1 \right) \quad (19.20)$$

The symbol $\tilde{E}$ is used to indicate that this is *not* a correctly formulated energy of a two-port storage element. However, it happens that this energy indeed is interpreted this way, which leads to the even more strange relations

$$\frac{\partial \tilde{E}(\lambda_1, i_1)}{\partial i_1} = \lambda_a \quad (19.21)$$

$$\frac{\partial \tilde{E}(\lambda_1, i_1)}{\partial \lambda_1} = i_1 \quad (19.22)$$

such that i_1 both plays the role of a magnetic effort and *seems* to play the role of a magnetic q -variable.

In this case the Legendre transform with respect to λ_1 is:

$$\tilde{E}^*(i_1, \lambda_a) = \frac{1}{2} L_1 i_1^2 + \lambda_a i_1 = E^*(i_1, i_2) - \frac{1}{2} L_2 i_2^2 \quad (19.23)$$

Figure 19.9d also shows that the use of $\tilde{E}^*(i_1, \lambda_a)$ means that the (infinitely) large self-co-energy of the field $\frac{1}{2} L_2 i_2^2$ is omitted, such that only the self-co-energy of the coil $\frac{1}{2} L_1 i_1^2$ and the mutual co-energy $\lambda_a i_1$ remain. However, the definition of λ_a requires that the (co-)energy is *not* mutual, such that the Se-source cannot be omitted in order to create an open (second) port of a two-port storage element. If an attempt is made to describe the behavior of the external field in another way than with an Se-source, then the analysis that uses the concept of the applied flux λ_a is not valid. Consequently, this model cannot be used in that case.

The additional insight that the current system approach offers is underlined once more by the simplicity of the model shown in figures 19.10e and 19.10f.

Summarizing it can be stated that, if an assumption made about the nature of the external field that leads to a simplified storage element *and* a source, this source cannot be considered to be separate from this model and the model becomes a one-port.

ASSIGNMENT 19.8

F is defined as the force that should be applied to a coil in order to keep it at a (positive) distance x (in axial direction) of a permanent magnet. How would the flux in the coil applied by the magnet behave as a function of this distance? What is the sign of the (partial) derivative of F with respect to x if λ_a is inversely proportional to x^2 ? What is the interpretation of can this sign?

If it is assumed in the sequel that the applied flux is inversely proportional with the square of the distance to the coil (the sign of the applied flux cannot depend on the sign of x), a two-port storage element is created with a magnetic and a mechanical port characterized by non-linear constitutive relations. An example of a practical application is a combination of a permanent magnet and a voice coil in a loudspeaker. Just like the membrane of the electret microphone (cf. the solution of assignment 18.9) the voice coil is elastically suspended (via the elastic suspension of the cone of the loudspeaker) in order to prevent intrinsic instability. The relation between the applied flux due to the permanent magnet and the distance x is assumed to be (analog to the solution of assignment 19.8)

$$\lambda_a = \frac{A}{x^2 + B} \quad (19.24)$$

If the contribution of the elastic suspension to the co-energy with respect to λ , $\Delta \tilde{E}^*(i, x)$, where

$$\Delta E_{\text{elastic}}(x) = \frac{K(x - x_0)^2}{2} = -\Delta \tilde{E}^*(i, x) \quad (19.25)$$

is added to (19.23) substitution of (19.24) results in

$$\tilde{E}^*(i, x) = \frac{L i^2}{2} + \frac{i A}{x^2 + B} - \frac{K(x - x_0)^2}{2} \quad (19.26)$$

where i is the current through the voice coil, L the self induction of the voice coil, K the spring constant of the suspension and x_0 the rest distance (in axial direction) of the voice coil with respect to the permanent magnet. This leads to the derivation of the constitutive relations for the flux linkage $\lambda(i, x)$ and the force $F(i, x)$. It is used that the difference between $\tilde{E}^*(i, x)$ and $E^*(i, x)$ depends neither on i nor on x , such that:

$$\lambda(i, x) = \frac{\partial E^*}{\partial i} = \frac{\partial \tilde{E}^*}{\partial i} = Li + \frac{A}{x^2 + B} = Li + \lambda_a \quad (19.27)$$

$$F(i, x) = \frac{-\partial E^*}{\partial x} = \frac{-\partial \tilde{E}^*}{\partial x} = \frac{2xiA}{(x^2 + B)^2} + K(x - x_0) \quad (19.28)$$

In the operating point $(i_0, x_0) = (0, x_0)$ the following linear approximation of (19.27) can be made:

$$\begin{aligned} \Delta\lambda(i, x) &= \lambda - \lambda_0 = \left. \frac{\partial\lambda}{\partial i} \right|_{i_0, x_0} (i - i_0) + \left. \frac{\partial\lambda}{\partial x} \right|_{i_0, x_0} (x - x_0) \\ &= Li - \frac{2x_0 A}{(x_0^2 + B)^2} (x - x_0) \end{aligned} \quad (19.29)$$

In a similar way (19.28) results for ΔF in:

$$\begin{aligned} \Delta F &= F - F_0 = F = \left. \frac{\partial F}{\partial i} \right|_{i_0, x_0} (i - i_0) + \left. \frac{\partial F}{\partial x} \right|_{i_0, x_0} (x - x_0) = \\ &= \frac{2x_0 A}{(x_0^2 + B)^2} i + \left(\frac{2Ai_0}{(x_0^2 + B)^2} - \frac{8Ax_0^2 i_0}{(x_0^2 + B)^3} + K \right) (x - x_0) = \\ &= \frac{2x_0 A}{(x_0^2 + B)^2} i + \left(2Ai_0 \frac{B - 3x_0^2}{(x_0^2 + B)^3} + K \right) (x - x_0) \end{aligned} \quad (19.30)$$

Because at the rest distance x_0 it holds that $F(x_0) = F_0 = 0$, it also holds that $\Delta F = F - F_0 = F$.

The equations (19.29) and (19.30) can be written in matrix form:

$$\begin{pmatrix} \lambda - \lambda_0 \\ F \end{pmatrix} = \begin{pmatrix} L & -\frac{2x_0 A}{(x_0^2 + B)^2} \\ \frac{2x_0 A}{(x_0^2 + B)^2} & 2Ai_0 \frac{B - 3x_0^2}{(x_0^2 + B)^3} + K \end{pmatrix} \begin{pmatrix} i \\ x - x_0 \end{pmatrix} \quad (19.31)$$

The lower right term of the matrix relates F and $(x - x_0)$. This means that it behaves like a spring constant. As it is not a ‘true’ spring constant it is called a *virtual spring constant*. Note that the constitutive matrix is not symmetric after Legendre transformation (with respect to λ), but that the moduli of the non-diagonal terms are equal.

A cycle process analog to the case of the two coupled coils, i.e. with a periodic signal, can be used here too to transduce electric power —supplied by an amplifier— transduction into mechanical power: the motion of the cone (figure 19.11). The loudspeaker consists of a permanent magnet connected to the housing. A coil rigidly connected to the cone can shift over the magnet. Note

that if the operating point is chosen such that $x_0 = \sqrt{\frac{B}{3}}$ the virtual spring

constant $2Ai_0 \frac{B - 3x_0^2}{(x_0^2 + B)^3} + K$ equals K and does not depend on i_0 accordingly,

which will improve the linearity of the transfer of the loudspeaker in principle.

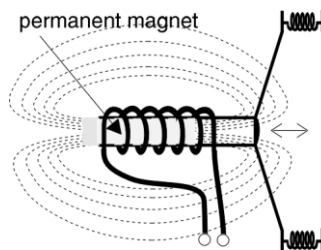


FIGURE 19.11

Loudspeaker, transduction of electric power in mechanical and acoustic power

If the cone is considered a kind of *piston* it becomes clear that it transfers the mechanical power in acoustic power that leaves the system in the form of sound waves.

Obviously, two coils can also be put into a configuration in which they can move with respect to each other, where a similar expression is found for the force. For example, this is applied in a so-called electrodynamicometer. In this device that can be used for accurate measurement of currents (with a relative error of $2 \cdot 10^{-5}$), four coils are connected in series through which the current to be measured is guided. The position of two coils is fixed and the two other coils are connected with the two sides of the lever of a balance (pair of scales) in such a way that these coils move in co-axial direction with the fixed coils and the forces are opposite. This causes a torque that can be compensated by weights on the balance. These weights are an accurate measure for the current through the coils. As the torque is proportional to the square of the current this current measurement device also works for AC.

In correspondence with the result found in section 2.4, viz. that an amount of work $-Mi_1i_2$ is required to bring two coils together, the addition of a magnet with applied flux λ_a to a coil requires an amount of work $-i_1\lambda_a$, where it was shown already that λ_a equals Mi_2 when this magnet is replaced by a coil. Note that a sign error would be made again if the co-energy would have been used instead of the energy and $i_1\lambda_a$ would have been as *energy* increase, because this would mean for the chosen orientations that the coil and the magnet would repel each other, which is not the case.

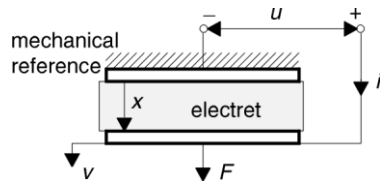
4 The electret microphone

After the linear example of the coupled coils and its (non-linear) extension into a coil in an external field, a cycle process with the non-linear example of the electrostatic transducer, the capacitor with moving plate (cf. chapter 18) is

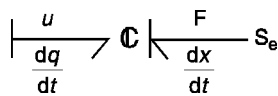
Electret

Electret
microphones

discussed. An elastic dielectric with spring constant K [N m^{-1}] is inserted between the plates in order to obtain an intrinsically stable system. It is immediately clear that this system cannot transduce a continuous, constant power, as this would mean that a linear motor or generator would be constructed in which the plates could be separated ‘infinitely’ far from each other. For such a configuration the storage elements *saturate* quickly. Chapter 18 already showed that a bias voltage is required. A simple way to realize this is the use of an *electret*. This material preserves its polarization after being polarized by application of a strong electric field (of the order of $1 \cdot 10^6 \text{ V m}^{-1}$). The material (mostly special kinds of wax, plastic or ceramic material) consists of molecules with a permanent polarization that are randomly oriented before application of the electric field, but are brought in line during polarization in the direction of the applied field and ‘freeze’ in this position. Sometimes the field is applied during solidification of the material. In order to use this phenomenon the material should be good isolator and hydrophobic, i.e. water repellent, because the adhesion of (conducting) water would minimize the effect. The phenomenon of the electret, discovered in 1925, is applied in electrostatic microphones that are called *electret microphones* accordingly. The bundles of characteristics of the ports of an electrostatic transducer will be sketched. Herein a cycle process will be indicated that could take place in an electrostatic transducer like the electret microphone. The previous chapter already introduced the derivation of the constitutive relations of the electrostatic transducer. The difference with the situation in the electret microphone is that an elastic dielectric or an elastic suspension of the membrane is present (figure 19.12).

FIGURE 19.12
Electret microphone

The force F equals zero when the distance between the plates has a certain value that has to be positive, e.g. $x_0 = d$ [m]. As the force at the microphone equals zero when air pressure variations are absent, this force should be considered the imposed (independent) variable. Consequently, the mechanical port does not have its preferred integral causality (figure 19.13).

FIGURE 19.13
Bond graph of the electret microphone

This means that stored energy should be Legendre transformed with respect to the position of the plate x , thus making F the independent variable. Pressure variations thus result in small force variations around this equilibrium. The equilibrium charge q_0 of the electret causes a force at the plates, such that the equilibrium distance between the plates is smaller than d . This will be encountered in the analysis of this two-port C model of the electret microphone.

The analysis will describe respectively: the energy $E(q,x)$, the voltage $u(q,x)$ and the force $F(q,x)$, the co-energy with respect to x , $E^*(q,F)$, and the expressions for $u(q,F)$ and $x(q,F)$ that can be derived from the co-energy. Next a linearization around the operating point ($q = q_0$, $F = 0$) is made, such that a linear two-port storage element results. The derivation of the cycle process is

then completely similar to that of the coupled coils (also a linear two-port storage element).

$$E(q, x) = \frac{q^2}{2C(x)} + \frac{K(x-d)^2}{2} = \frac{q^2 x}{2A\epsilon} + \frac{K(x-d)^2}{2} \quad (19.32)$$

$$F = \frac{\partial E}{\partial x} = \frac{q^2}{2A\epsilon} + K(x-d) \quad (19.33)$$

such that

$$x = \frac{F}{K} - \frac{q^2}{2A\epsilon K} + d \quad (19.34)$$

In the first step of the following derivation of the co-energy with respect to x only the term $(x-d)$ is substituted using (19.34) in order to be able to recognize the form of the Legendre transform. In the second step all other variables x are substituted and simplified in the third step:

$$\begin{aligned} E^*(q, F) &= Fx - \left(\frac{q^2 x}{2A\epsilon} + K \frac{\left(\frac{F}{K} - \frac{q^2}{2A\epsilon K} \right)^2}{2} \right) = \\ &= F \left[\left(\frac{F}{K} - \frac{q^2}{2A\epsilon K} \right) + d \right] - \left[\left(\frac{F}{K} - \frac{q^2}{2A\epsilon K} \right) + d \right] \frac{q^2}{2A\epsilon} - \left[\frac{F^2}{2K} - \frac{Fq^2}{2A\epsilon K} + \frac{\left(\frac{q^2}{2A\epsilon} \right)^2}{2K} \right] = \\ &= \frac{F^2}{2K} - \frac{Fq^2}{2A\epsilon K} + Fd + \frac{q^4}{8KA^2\epsilon^2} - \frac{q^2 d}{2A\epsilon} \end{aligned} \quad (19.35)$$

Partial differentiation is used to check the result (cf. (19.34))

$$x(q, F) = \frac{\partial E^*}{\partial F} = \frac{F}{K} - \frac{q^2}{2A\epsilon K} + d \quad (19.36)$$

but also

$$\begin{aligned} u(q, F) &= \frac{-\partial E^*}{\partial q} = \frac{Fq}{A\epsilon K} - \frac{q^3}{2KA^2\epsilon^2} + \frac{qd}{A\epsilon} = \\ &= \frac{q}{A\epsilon} \left(\frac{F}{K} + d - \frac{q^2}{2A\epsilon K} \right) \end{aligned} \quad (19.37)$$

$$u(q = q_0, F = 0) = \frac{q_0}{A\varepsilon} \left(d - \frac{q_0^2}{2A\varepsilon K} \right) = u_0, \quad \text{the bias voltage} \quad (19.38)$$

Note that the partial derivative(s) with respect to the variable(s) that have been Legendre transformed do not have a minus (–) sign, but the partial derivative(s) of the variable(s) that have not been Legendre transformed do, so:

$$\begin{aligned} \frac{\partial E^*(q, F)}{\partial q} &= -u(q, F) \\ \frac{\partial E^*(q, F)}{\partial F} &= +x(q, F) \end{aligned}$$

From (19.38) can be derived that the distance between the plates in case $F = 0$ is the equilibrium distance (rest distance) d corrected with the displacement

$$-\frac{q_0^2}{2A\varepsilon K}$$

that causes the attraction between the plates for a charge q_0 .

If the characteristics corresponding to (19.36) in the x – F plane and corresponding to (19.37) in the u – q plane respectively are sketched, then the u – q plane shows lines through the origin of which the slope in the origin is determined by F in such a way that that this slope increases for increasing values of F . Further away from the origin the curve will reach a maximum and bend back next towards the q –axis (figure 19.14).

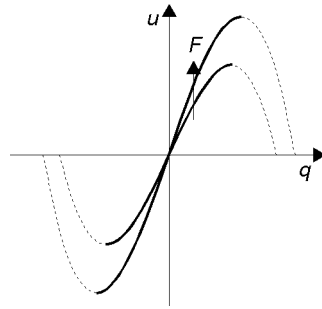


FIGURE 19.14
Characteristics in the u – q plane

This means that the storage element is not intrinsically stable beyond this maximum, which should result (according to this model) in a collapse and short-circuit of the plates. In fact this maximum indicates the situation where the force at the plates overcomes the opposing force of the elastic dielectric, which could be interpreted as a collapse of the plates. However, in practice the electret material will not behave at this point according to *Hooke's law* (i.e. linearly, cf. chapter 2), which means that the solution (and thus the model) found is not realistic anymore. Consequently, the value of q_0 will have to be amply below the value of q where u reaches a maximum. A similar constraint holds for the straight characteristics in the x – F plane that all have a slope $1/K$, but do not intersect with the origin: in case $q = 0$ the characteristic intersects the vertical axis in $x = d$, the remaining points of intersection are below this point. The characteristic for the maximum value of q intersects the vertical axis still above

the origin. The characteristics are only valid for positive values of x , so above the horizontal axis (figure 19.15).

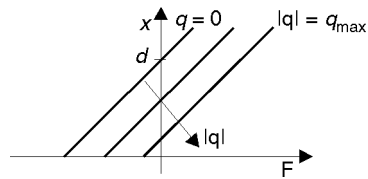


FIGURE 19.15
Characteristics in the x - F plane

The microphone will only result in vibrations of the plate, i.e. small variations around an equilibrium. This equilibrium lies around q_0 —the rest charge of the electret— and around $d - q_0^2/2A\epsilon K$ —the distance between the plates at $F = 0$.

The linear constitutive relations follow from the partial derivatives of u and x with respect to q and F that can be written in matrix form:

$$\begin{pmatrix} u - u_0 \\ x - x_0 \end{pmatrix} = \begin{pmatrix} \frac{-3q_0^2}{2KA^2\epsilon^2} + \frac{d}{A\epsilon} & \frac{+q_0}{A\epsilon K} \\ \frac{-q_0}{A\epsilon K} & 1/K \end{pmatrix} \begin{pmatrix} q - q_0 \\ F \end{pmatrix} \quad (19.39)$$

The cycle process can take the form sketched in figure 19.16 (obtained by means of simulation). Note that the constitutive matrix (19.39) is not anymore symmetric after Legendre transformation, but again has equal moduli of the non-diagonal terms.

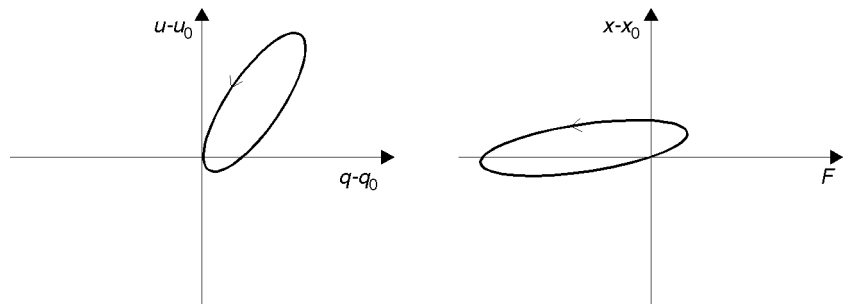


FIGURE 19.16 Cycle processes in the u - q plane and in the x - F plane

The decomposition of a two-port storage element with mixed causality in section 2.3 immediately leads to the decomposition based on (19.39) shown in figure 19.17a.

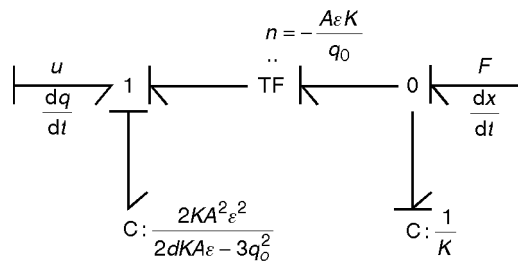


FIGURE 19.17a Decomposition of the two-port storage element representing a model of an electret microphone

This figure can be simplified into figure 19.17b by changing the positive orientation of the TF (half arrow direction) and by using

$$x_0 = d - \frac{q_0^2}{2A\epsilon K}$$

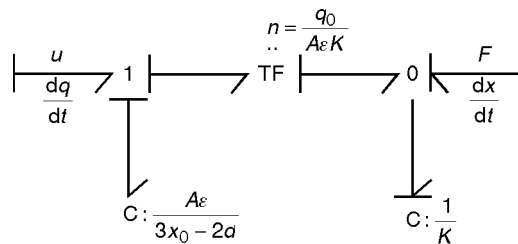


FIGURE 19.17b Simplification of figure 19.17a

This decomposition immediately shows that coupling disappears if $q_0 = 0$ —in that case there are two separate one-port storage elements, where the electric capacitance equals $A\epsilon/d$ and the mechanical spring constant equals K —and that it should hold that $x_0 > 2d/3$ in order to maintain an intrinsically stable storage element.

5 Coil with moving core

A coil with moving, soft iron core is a storage element with a magnetic and a mechanical port just like the elastic coil (cf. figure 19.18).

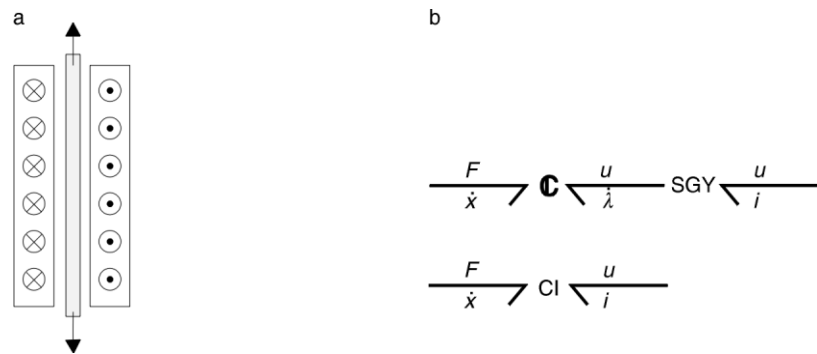


FIGURE 19.18 Coil with moving core
 a cross section of the coil
 b two possible bond graphs (core assumed massless)

A good starting point for the analysis of this model is to establish how the self induction of the coil depends on the position of the core: $L = L(x)$. If $x = 0$ is chosen to be the position where the core is in the middle of the coil in such a way that L reaches a maximum ($L(0) = L_{\max}$), then $L(x)$ will be a bell-shaped function that for large values of x —for example for $|x| > |x_{\max}|$ —approaches the self induction of the empty coil $L_{\min}$ (figure 19.19).

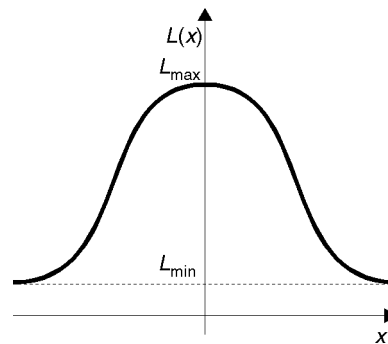


FIGURE 19.19
 Relation between $L(x)$ and x

In this case x has an upper bound $x_{\max}$ for which the storage element still is a two-port. Although this means that this transducer cannot be constructed such that it becomes a linear transducer (linear motor or generator), it can be used again to design a power converting cycle process. In this case x can become negative, in contrast to the electro- and magnetomechanical transducers encountered so far.

If the bell-shaped function $L(x)$ should switch smoothly into $L_{\min}$, the derivative of $L(x)$, $L'(x)$, should have at least three zero crossings, viz. one in $L_{\max}$ and two when $L_{\min}$ is reached. This means that $L'(x)$ should be at least a third order polynomial in x . In turn, this means that this part of $L(x)$ should be described by a polynomial in x that is at least of the fourth order, *for example* (higher order polynomials and other functions are possible too),

$$L(x) = ax^4 + bx^3 + cx^2 + dx + e \quad (19.40)$$

The coefficients a , b , c , d and e are expressed in the known parameters. The value for e follows directly from $L(0) = L_{\max} = e$. Based on the fact that L has a maximum in $x = 0$

$$\left. \frac{dL(x)}{dx} \right|_{x=0} = L'(0) = 4ax^3 + 3bx^2 + 2cx + d = d = 0 \quad (19.41)$$

and because $L(x)$ should approach $L_{\min}$ smoothly at $|x| = x_{\max}$ it should hold that

$$\begin{aligned} L'(x_{\max}) &= 4ax_{\max}^3 + 3bx_{\max}^2 + 2cx_{\max} = 0 \\ &\rightarrow 4ax_{\max}^2 + 3bx_{\max} + 2c = 0 \end{aligned} \quad (19.42)$$

$$\begin{aligned} L'(-x_{\max}) &= -4ax_{\max}^3 + 3bx_{\max}^2 - 2cx_{\max} = 0 \\ &\rightarrow -4ax_{\max}^2 + 3bx_{\max} - 2c = 0 \end{aligned} \quad (19.43)$$

such that $b = 0$ and $4ax_{\max}^2 + 2c = 0$. The smooth transition also requires that

$$L(x_{\max}) = L_{\min} = L(-x_{\max}) \quad (19.44)$$

such that

$$L(|x_{\max}|) = ax_{\max}^4 + cx_{\max}^2 + L_{\max} = L_{\min} \quad (19.45)$$

Combination with $c = -2ax_{\max}^2$ results in:

$$a = \frac{L_{\max} - L_{\min}}{x_{\max}^4} \quad \text{and} \quad c = -\frac{2(L_{\max} - L_{\min})}{x_{\max}^2} \quad (19.46)$$

while it was already found that

$$\begin{aligned} b &= 0 \\ d &= 0 \\ e &= L_{\max} \end{aligned}$$

such that with $X = x/x_{\max}$

$$L(X) = X^4(L_{\max} - L_{\min}) - 2X^2(L_{\max} - L_{\min}) + L_{\max} \quad (19.47)$$

Note well that this is just a possible choice for $L(x)$, not some physical ‘law’

This result can be used to derive $F(x, i)$ from the co-energy

$$E^* = \frac{L(x)i^2}{2}$$

$F(x, i)$ is the force that should be applied to the core in order to achieve a displacement x when a current i is running through the coil.

$$\begin{aligned}
 F(x, i) &= \frac{-\partial E^*}{\partial x} = \frac{-\partial}{\partial x} \frac{L(x)i^2}{2} = -\frac{i^2}{2} \frac{dL(X)}{dX} = -\frac{i^2}{2} \frac{dL(X)}{dX} \frac{dX}{dx} = \\
 &= (L_{\max} - L_{\min}) \frac{2i^2}{x_{\max}} (1 - X^2) X = \\
 &= \frac{2i^2 x}{x_{\max}^4} (x_{\max}^2 - x^2) (L_{\max} - L_{\min})
 \end{aligned} \tag{19.48}$$

ASSIGNMENT 19.9

Compute the area of intrinsic stability of this two-port storage element if $i \neq 0$.

LVDT

In contrast to the electromechanical and magnetomechanical two-ports encountered so far this two-port is intrinsically stable around $x = 0$ without the necessity of an additional mechanical stiffness. This makes this transducer perfectly suited as a linear position sensor. Such a position sensor is called an *LVDT* (Linear Variable Displacement Transmitter). If the core is connected to the moving part the measurement has the advantage of being free of contact.

Linearization around the operating point (i_0, X_0) results in the following constitutive relations in matrix form:

$$\begin{pmatrix} F - F_0 \\ \lambda - \lambda_0 \end{pmatrix} = (L_{\max} - L_{\min}) \begin{pmatrix} \frac{2i_0^2(1-3X_0^2)}{x_{\max}^2} & \frac{4i_0(1-X_0^2)X_0}{x_{\max}} \\ -\frac{4i_0(1-X_0^2)X_0}{x_{\max}} & X_0^2(X_0^2-2) + \frac{L_{\max}}{(L_{\max} - L_{\min})} \end{pmatrix} \begin{pmatrix} x - x_0 \\ i - i_0 \end{pmatrix} \tag{19.49}$$

According to the result in section 2.3 the coil with the moving core can be represented after linearization by the decomposition in figure 19.20. The electric port does not have preferred causality in this case.

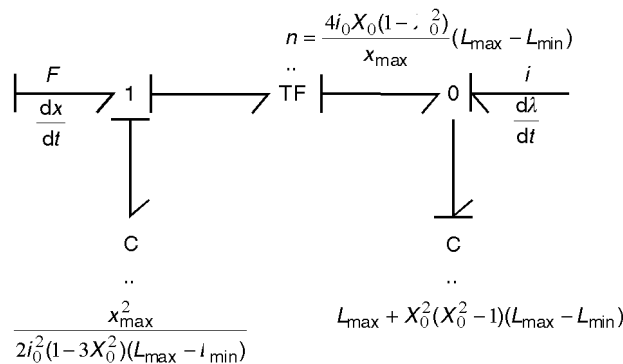


FIGURE 19.20

Decomposition of the coil with moving core

The decomposition shows the effect of the decrease of the difference between $L_{\max}$ and $L_{\min}$: the coupling decreases and the virtual spring constant of the mechanical port decreases. It also shows that the coupling disappears in $x_0 = 0$, although the two-port remains intrinsically stable there, while the virtual spring constant remains positive. When there is no current in the coil ($i_0 = 0$) both the coupling and the spring constant equals zero. This model result is realistic.

6 Cycle processes with gasses and liquids

6.1 ENGINEERING APPROXIMATIONS OF THE CARNOT CYCLE

Chapter 17 introduced a number of cycle processes with a medium in gas phase or liquid phase. These are all engineering approximations of the Carnot cycle. The Carnot cycle can be represented in a bond graph by a two-port C for example (cf. figure 19.21).

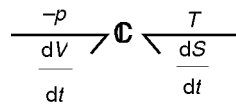


FIGURE 19.21
Bond graph of a Carnot cycle

The causality of this two-port C often depends on the segment of the cycle. The ports do not always get their preferred causality in the Carnot cycle. This means that engineering approximations of these parts of the cycle are not easily achieved. The causality is indicated for all the cycles to be treated herein. As discussed in chapter 7 a differential causality in principle allows an instantaneous change of a state and consequently an infinitely large value of a flow or an effort (time derivative of the concerning state). However, in the cases below the state is only kept constant and not changed instantaneously. This can be interpreted as follows: the environment should act in that case as a very large storage element (that can be considered a source with respect to the two-port storage element accordingly) that can impose the state. In practise always some effort difference (or gradient) will occur with a resistance effect ‘in between’ (figure 19.22a).

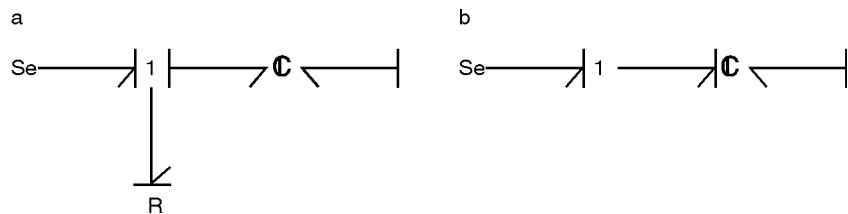


FIGURE 19.22 If the very small internal resistance of the source in a is omitted, one of the ports gets differential causality (b)

This resistance enables preferred integral causality for both storing ports. However, if the process is realized such that this resistance is as small as possible the time constant (note the causal path between the R-port and the C-port) becomes negligibly small. The model in figure 19.22b, where the port of the two-port storage element obtains a differential causality, approximates this situation well. This is why in this particular case the process represented in figure 19.22b may be considered to have the desired causality.

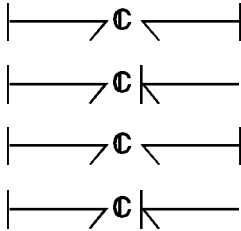
The definitions of the port variables (mechanical and thermal port) are in the following discussion of a number of engineering approximations of a Carnot cycle analog to figure 19.21, i.e. mechanical port at the left-hand side and thermal port at the right-hand side. In the Carnot cycle and in the other cycles the following terminology is encountered, which will be used here too to describe the processes:

Adiabatic

adiabatic or isentropic = with constant entropy

*Isothermal**Isobaric**Isochoric**isothermal* = with constant temperature*isobaric* = with constant pressure*isochoric* = with constant volume.

For reasons of completeness the parts of the Carnot cycle are repeated:



$A \rightarrow B$: $dV < 0$; $dS = 0$ (adiabatic compression)

$B \rightarrow C$: $dV > 0$; $dT = 0$ (isothermal expansion)

$C \rightarrow D$: $dV > 0$; $dS = 0$ (adiabatic expansion)

$D \rightarrow A$: $dV < 0$; $dT = 0$ (isothermal compression)

This means preferred integral causality at the $(p, -dV/dt)$ -port and alternating preferred integral causality and differential causality at the $(T, dS/dt)$ -port. Differential causality indicates that a temperature is imposed. However, this does not go beyond keeping this temperature constant. This means that no infinitely large entropy flows are needed. When the volume changes would be so fast that the entropy flows that are needed to keep the temperature constant would become so large that the irreversible effects cannot be neglected, then the model would need an extension by means of a resistor. This would recover the integral causality of the thermal port, such that there is no pure Carnot cycle left. Hence the constraint of the Carnot cycle is that the (volume) changes are slow, i.e. quasi-static.

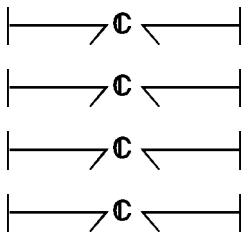
The engineering approximations of the Carnot cycle are summarized together with the corresponding causal bond graph(s). In all cases the ports are the $(p, -dV/dt)$ -port or mechanical port and the $(T, dS/dt)$ -port or thermal port.

ASSIGNMENT 19.10

Sketch the cycles described next in the p - V -plane and the T - S -plane respectively, assuming that the gas is ideal. This assignment is given beforehand, such that the reader can work on it while reading the following text.

Otto cycle

The German engineer *Nikolaus August Otto* (1832–1891) invented in 1861 the combustion engine (gasoline engine), a cyclic process that was the first attempt to approximate the Carnot cycle. This cycle is called *Otto cycle* accordingly:



$A \rightarrow B$: $dV = 0$; $dS > 0$ (isochoric heating: ‘explosion’)

$B \rightarrow C$: $dV > 0$; $dS = 0$ (adiabatic expansion)

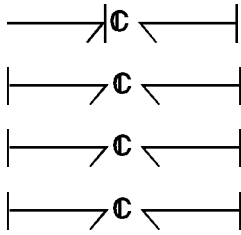
$C \rightarrow D$: $dV = 0$; $dS < 0$ (isochoric cooling)

$D \rightarrow A$: $dV < 0$; $dS = 0$ (adiabatic compression)

This means that one of the first cycles realized in practice has preferred causality at both ports during the whole cycle.

Diesel cycle

In 1890 another German engineer, *Rudolf Christian Karl Diesel* (1858–1913) invented an engine that was patented in 1892. This motor excelled in simplicity of the design and the high efficiency compared to the Otto cycle. The cycle in this process is called *Diesel cycle* accordingly:



$A \rightarrow B$: $dp = 0$; $dS > 0$ (isobaric heating and expansion: combustion)

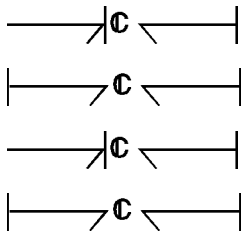
$B \rightarrow C$: $dV > 0$; $dS = 0$ (adiabatic expansion)

$C \rightarrow D$: $dV = 0$; $dS < 0$ (isochoric cooling)

$D \rightarrow A$: $dV < 0$; $dS = 0$ (adiabatic compression)

Only during the first segment, the combustion phase, the $(p, -dV/dt)$ -port has differential causality. This is approximated in practice by injecting the fuel at the end of the compression phase, i.e. under high pressure and temperature, in the combustion chamber and to (self-)ignite it. The fuel is injected in such a way that the pressure remains approximately constant during expansion. This requires a certain control of the process to realize this segment of the Diesel cycle in practice.

Joule–Brayton cycle



$A \rightarrow B$: $dp = 0$; $dS > 0$ (isobaric heating)

$B \rightarrow C$: $dV > 0$; $dS = 0$ (adiabatic expansion)

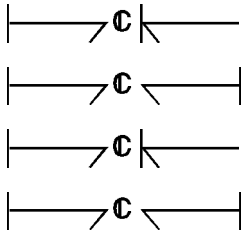
$C \rightarrow D$: $dp = 0$; $dS < 0$ (isobaric cooling)

$D \rightarrow A$: $dV < 0$; $dS = 0$ (adiabatic compression)

During the first and the third segment the $(p, -dV/dt)$ -port has differential causality. In gas turbines a continuous process realizes this in a so-called open cycle, to be discussed in the next section. This means that the air leaving the turbine is not taken in immediately again, but that fresh air is taken in. Hence, the isobaric cooling is not a true cooling of (the same) air, but the intake of (other) air with a constant ambient temperature. The environment thus serves as a very large storage element that can be considered both a temperature source and a pressure source compared to the storage in the turbine. The pressure remains also constant during combustion, because the process is continuous. This means that during a constant heat supply to the combustion chamber a more or less constant pressure is achieved.

Stirling cycle

The Scottish preacher and inventor *Robert Stirling* (1790–1878) invented a cycle in which the air used is always the same, such that there are no exhaust gases. The closed chamber, in which the air is trapped, is closed at the outside by two pistons and contains a third freely moving piston that moves the air between a heated chamber and a cooled chamber. In that way isochoric heating and cooling can be achieved. The Stirling engine has been studied by Philips Eindhoven since 1938. Since 1958 General Motors also contributed to the research. It appeared possible to build engines with a power of 4000 kW. It is not completely clear what the pitfall (technological, economical, safety, political or a combination of this factors) prevented large-scale application of this engine. It may be that the potential positive environmental aspects of this engine can still change this. This engine is based on the so-called *Stirling cycle*:



$A \rightarrow B$: $dV > 0$; $dT = 0$ (isothermal expansion, i.e. simultaneous heating)

$B \rightarrow C$: $dV = 0$; $dS < 0$ (isochoric cooling)

$C \rightarrow D$: $dV < 0$; $dT = 0$ (isothermal compression, i.e. simultaneous cooling)

$D \rightarrow A$: $dV = 0$; $dS > 0$ (isochoric heating)

During the first and third segment the $(T, dS/dt)$ -port has differential causality. This means that it is assumed that the medium that is moved between the hot and the cold chamber instantaneously has the temperature of that chamber. This means that construction of the Stirling engine should be focused on optimizing the heat conduction from and to the medium. In practice, when losses cannot be neglected, there will always be some a heat resistance between the heat exchanger in the chamber and the medium itself, which turns the differential causality of the port in fact again into an integral causality.

Sometimes, for example in a Rankine cycle, the two-phase area of a medium is used to increase the enclosed area of the cycle, i.e. the amount of transduced energy per cycle. This makes the different segments of the cycle more complex, which is beyond the context of this text.

6.2 CYCLE PROCESS WITH CYCLING WORKING FLUID

In typical engineering cycle processes—for example in cooling and heating systems and electric power plants (fired by coal, by gas or by nuclear fuel)—a fluid medium in liquid or gas phase called *working fluid* is present which goes through a cycle process as described in section 6.1. However, this cycle process is not achieved by varying the boundary conditions of the system in time like in the case of the transformer or the cylinder–piston combination filled with gas, but by guiding the working fluid through a system where it virtually encounters the different boundary conditions that can remain constant (in time) themselves, such that the cycling working fluid ‘sees’ them vary in time during circulation. In other words: the required changes in time of the working fluid (medium) are mapped on space. By guiding the working fluid (medium) through that space the intended cycle is realized. As this working fluid usually cycles in a closed loop, there is not only a cycle in the state space, but also a tangible spatial cycle, which sometimes causes the misunderstanding that the name ‘cycle process’ is constrained to the latter type of systems. In contrast with the examples without a working fluid the power transduction in such a system between the ports with the environment that are of interest can be indeed continuous.

In case air is used as a working fluid the cycle is often open, where cold air is taken in from the environment and hot air is returned to the environment, usually convecting exhaust gasses, while it sometimes preheats the intake air via a heat exchanger. An example of such an open cycle is found in a gas turbine.

ASSIGNMENT 19.11

When you do not know already, find out in an (engineering) encyclopedia how the cooling cycle in your refrigerator operates. Do you have any idea why in the past, before the effects on the Ozone layer were known, Freon gas was chosen as a working fluid?

7 A domain independent cycle process

7.1 BASIC EQUATIONS

The analogy between the examples treated so far leads to the obvious conclusion that a two-port C can be used to transduce power via a cycle process independent of the domains of the ports. If it is assumed that the energy can be written as a function of the stored quantities, then the constitutive relations $e_1(q)$ and $e_2(q)$ can always be found by partial differentiation. It is possible to linearize around an operating point by repeated partial differentiation of these relations. The result shows immediately whether the two-port storage element is intrinsically stable in that operating point: intrinsic stability requires a positive-definite matrix and positive terms at the diagonal. It is important to make a clear distinction between the energy and the different forms of co-energy that may be used to characterize the two-port.

Next a cycle process can be designed by imposing some periodic signal *either* to the flows and thus to the storage element contents *or* to the equilibrium determining variables. In the latter case a causal inversion of the concerning port is required that corresponds to a Legendre transform of the energy to the storage element content of the concerning port, as demonstrated already in a number of examples. In all cases the periodic nature of the port variables results in a cycle process that can be represented for example in the bundles of characteristics of the ports (e - q -planes). These bundles of characteristics are, in particular in the non-linear case, a good starting point for the design of a cycle process, viz. the choice of the operating point and the periodic signals at the ports.

7.2 DESIGN OF CYCLE PROCESSES

After gaining a domain independent knowledge of cycle processes and thus of power transduction, it can be used to define hypothetical cycle processes. This requires the choice of the domains of the two ports and a consideration how such a two-port storage element can be approximated in practice.

It is nice – and insightful – to check all possible domain combinations. A number of them has already been treated extensively, in particular almost all combinations with a linear displacement. It requires little imagination to add examples where the angular displacement is the second state. A frequently used application is the rotating electric motor. Due to the cyclic nature of the angular displacement variable itself a continuous angular velocity corresponds to a cycle process: after 2π radians the previous state is recovered. However, at the magnetic port the cycle should be constructed, either by means of AC-signals, or by means of a commutator, a mechanical switch, often realized by means of rotating contacts and (carbon) brushes that convert a DC-signal into a periodic signal. As a consequence the power can never be completely continuous, although increasing the number of coils can approximate this. It goes beyond the scope of this text to treat all different practical realizations of electric motors. This is a discipline by itself. However, the principle of transduction fits in the presented system approach, such that a motor model can be implemented in a consistent way in the system model if this discipline brings up the proper constitutive relations. This enables straightforward simulation of the dynamic behavior including both electromagnetic and mechanical phenomena. Nevertheless, analog to the model of the electric transformer a generic linear bond graph model of a simple DC-motor can be derived from a (co-)energy.

Starting from this model the consequences of neglecting for example the storage of magnetic energy or adding the kinetic energy of the rotor can be analyzed.

7.3 THE CYCLE PROCESS IN THE ELECTRIC MOTOR

The principle of the electric motor is based on the dependence of the stored magnetic energy on the configuration of coils. These are both the rotor coils and the stator coils that can rotate with respect to each other or coils that can rotate either with respect to a magnetic field or with respect to magnetic material (magnets). In other words: the magnetic energy that is stored in a motor is not only a function of the flux linkage λ of a coil, but also of the angular position φ of the rotor with respect to the stator:

$$E_{\text{magnetic}} = E(\lambda, \varphi) \quad (19.50)$$

If this energy is known, then a two-port storage element model can be taken as a starting point for the motor and the motor current i can be found with

$$i = \frac{\partial E(\lambda, \varphi)}{\partial \lambda} \quad (19.51)$$

and the motor torque T with

$$T = \frac{\partial E(\lambda, \varphi)}{\partial \varphi} \quad (19.52)$$

The complete energy is commonly not known, but only the cross term that depends on the angular position φ . This results from the magnetic interaction — mostly in the form of a co-energy term $E_{\text{interaction}}^*(i, \varphi)$, a Legendre transform with respect to λ . The torque that should be applied between both magnetic parts (rotor and stator) in order to maintain a specific angular position can be derived from this cross term as follows (the rest of the co-energy is independent of φ and does not contribute to this derivative)

$$T = -\frac{\partial E^*(i, \varphi)}{\partial \varphi} = -\frac{\partial E_{\text{interaction}}^*(i, \varphi)}{\partial \varphi} \quad (19.53)$$

Figure 19.23 shows the common sketch of the simplest possible DC-motor configuration, consisting of a rectangular winding (rotor) and a permanent magnet (stator).

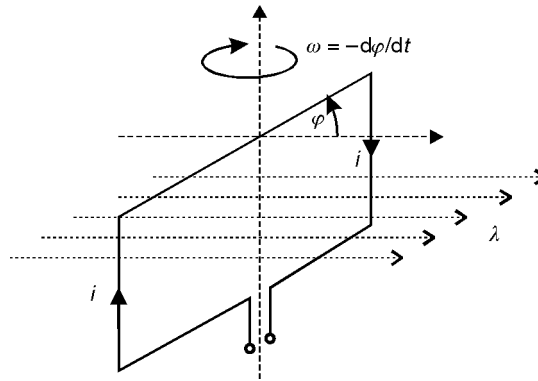


FIGURE 19.23 Sketch of an elementary electric motor

Note that chosen orientation implies that an angular displacement between 0 and 90 degrees means that the projection of the flux in the coil at the direction of the flux of the magnet is opposite to the direction of the flux of the magnet. This means that the cross term in the co-energy becomes negative and that work has to be performed in order to obtain this configuration. In the sketched case it holds that the applied flux by the permanent magnet depends on φ according to

$$\lambda_a(\varphi) = \lambda_{a\max} \sin \varphi \quad (19.54)$$

The cross term in the co-energy then becomes

$$E_{\text{interaction}}^*(i, \varphi) = -i\lambda_a(\varphi) \quad (19.55)$$

Consequently, the torque T to be applied equals

$$T(i, \varphi) = -\frac{\partial}{\partial \varphi} E_{\text{interaction}}^*(i, \varphi) = i \frac{\partial \lambda_a(\varphi)}{\partial \varphi} = i\lambda_{a\max} \frac{\partial \sin \varphi}{\partial \varphi} = i\lambda_{a\max} \cos \varphi \quad (19.56)$$

This means that if all given variables in the situation sketched in figure 19.23 are positive, the *applied* torque at the rotor should be negative and the motor will turn in negative direction, i.e. clockwise. This model result is realistic and corresponds to the analysis in chapter 3 that is based on the Lorentz force. The configuration in figure 19.23 only operates as motor if it equipped with a commutator that switches the direction (sign) of the current at $\varphi = 90^\circ + n \cdot 180^\circ$, where n is an integer number. In formula form

The sgn-function (for signum = sign) returns the sign of its argument.

$$i = i_0 \operatorname{sgn}(\cos \varphi) \quad (19.57)$$

A commutator often consists of copper (brass) contacts at the rotor on which carbon sticks are pressed by an elastic suspension at the stator. The common terminology for these sticks is 'carbon brushes' or 'graphite brushes' and has an historic background: the first electric motors were equipped with brushes of copper wires. Nowadays 'brushless' DC-motors are frequently used, in which the commutation is achieved by means of semiconductor switches (thyristors).

Due to the commutation the torque T to be applied maintains its sign at $i_0\lambda_{a\max}$:

$$\begin{aligned}
 T(i, \varphi) &= i_0 \operatorname{sgn}(\cos \varphi) \lambda_{a\max} \cos \varphi = \\
 &= i_0 \lambda_{a\max} \operatorname{sgn}(\cos \varphi) [|\cos \varphi| \operatorname{sgn}(\cos \varphi)] = i_0 \lambda_{a\max} |\cos \varphi|
 \end{aligned} \tag{19.58}$$

This causes the opposite applied torque also to keep its sign, such that the rotor keeps turning.

Although a three-port storage element with 2 magnetic and 1 rotation-mechanical port is considered, similar assumptions are made as in the case of the coil in the permanent magnetic field in section 3. This section showed that the derivation of the constitutive relations of the remaining two-port storage element only requires the self-co-energy of the coil and the already mentioned interaction co-energy, because the remaining co-energy term, the self-co-energy of the magnet, does not depend on i or φ . This co-energy is expressed in i_0 and φ (the cross term is negative due to the opposite direction of the fluxes):

$$\begin{aligned}
 \tilde{E}^*(i_0, \varphi) &= \frac{1}{2} L (i_0 \operatorname{sgn}(\cos \varphi))^2 - i_0 \operatorname{sgn}(\cos \varphi) \lambda_{a\max} \sin \varphi = \\
 &= \frac{1}{2} L i_0^2 - i_0 \operatorname{sgn}(\cos \varphi) \lambda_{a\max} \sin \varphi
 \end{aligned} \tag{19.59}$$

Partial differentiation results again in expressions for T and λ :

$$T(i_0, \varphi) = -\frac{\partial}{\partial \varphi} \tilde{E}^*(i_0, \varphi) = i_0 \operatorname{sgn}(\cos \varphi) \lambda_{a\max} \cos \varphi = i_0 \lambda_{a\max} |\cos \varphi| \tag{19.60}$$

$$\lambda(i_0, \varphi) = \frac{\partial}{\partial i_0} \tilde{E}^*(i_0, \varphi) = L i_0 - \operatorname{sgn}(\cos \varphi) \lambda_{a\max} \sin \varphi \tag{19.61}$$

The intrinsic stability of this two-port storage element is checked by partial differentiation:

$$\frac{\partial}{\partial \varphi} T(i_0, \varphi) = -i_0 \lambda_{a\max} \operatorname{sgn}(\cos \varphi) \sin \varphi \tag{19.62a}$$

$$\frac{\partial}{\partial i_0} T(i_0, \varphi) = \lambda_{a\max} |\cos \varphi| \tag{19.62b}$$

$$\frac{\partial}{\partial i_0} \lambda(i_0, \varphi) = L \tag{19.62c}$$

$$\frac{\partial}{\partial \varphi} \lambda(i_0, \varphi) = -\operatorname{sgn}(\cos \varphi) \lambda_{a\max} \cos \varphi = -\lambda_{a\max} |\cos \varphi| = -\frac{\partial}{\partial i_0} T(i_0, \varphi) \tag{19.62d}$$

The difference in sign between the mixed derivatives, as encountered earlier in (19.8), is a consequence of the mixed causality of the two-port. Equation (19.62a) shows that the system is only intrinsically stable in case that $-90^\circ + n \cdot 180^\circ < \varphi < 0^\circ + n \cdot 180^\circ$.

A number of these partial derivatives depends on φ . For variations in i_0 and φ that take place at a much larger time scale than the revolution time of the rotor—in other words: for signal frequencies that are much lower than the angular frequency of the motor—an average value of these partial derivatives can be determined per half round. The average value of the other half round is equal due to the signum-function that is a consequence of the commutator. This restores the state of the beginning of the half period, such that the cycle process is completed:

$$\begin{aligned} \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\partial}{\partial \varphi} T(i_0, \varphi) d\varphi &= \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} -i_0 \lambda_{a\max} \operatorname{sgn}(\cos \varphi) \sin \varphi d\varphi = \\ &= \frac{-i_0 \lambda_{a\max} \operatorname{sgn}(\cos \varphi)}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin \varphi d\varphi = \frac{-i_0 \lambda_{a\max} \operatorname{sgn}(\cos \varphi)}{\pi} \left[-\cos \varphi \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 0 \end{aligned} \quad (19.63)$$

$$\begin{aligned} \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\partial}{\partial i_0} T(i_0, \varphi) d\varphi &= \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \lambda_{a\max} |\cos \varphi| d\varphi = \frac{\lambda_{a\max}}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \varphi d\varphi = \\ &= \frac{\lambda_{a\max}}{\pi} \left[\sin \varphi \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\lambda_{a\max}}{\pi} [1 - (-1)] = \frac{2\lambda_{a\max}}{\pi} = 0.64 \lambda_{a\max} = K_T \end{aligned} \quad (19.64)$$

$$\frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\partial}{\partial \varphi} \lambda(i_0, \varphi) d\varphi = -\frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\partial}{\partial i_0} T(i_0, \varphi) d\varphi = -0.64 \lambda_{a\max} = -K_u \quad (19.65)$$

These relations represent a linear two-port storage element. Note that

- the first results equals zero, such that the corresponding storage element in the decomposition becomes a source. However, the value of this source should be zero, because a current less motor does not deliver a torque. As this zero-valued Se-source is connected to a 1-junction, it can be omitted. The fact that only one one-port storage element remains, is not caused by a full coupling, as was the case for the two coupled coils, but by the averaging process over (half) a period. In this particular case the position dependent coupling is essential for the transduction.
- a storage element at the mechanical side could obstruct the operation of the motor, because it would operate as a rotation spring that is wound by the motor.
- this model is only valued under the condition in which the averaging process is useful, viz. that $|\omega - \bar{\omega}| \ll \bar{\omega}$, where $\bar{\omega}$ is the average angular frequency.
- the motor turns against the positive direction of $d\varphi/dt$. If it is desired to represent the delivered mechanical power positively which requires an

inversion of the half arrow direction by means of a 0-junction, also a positive angular velocity $\omega = -d\varphi/dt$ is obtained.

Substitution of the obtained average values in the decomposition of a two-port storage element as derived at the end of section 2.3, results in the bond graph in figure 19.24a, taking the latter remark into account.

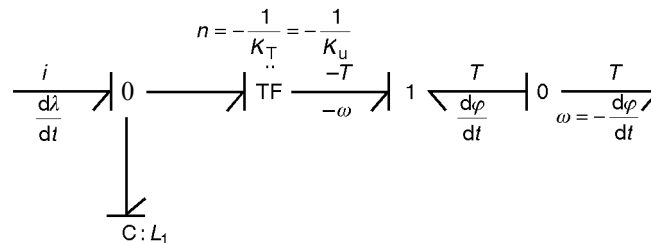


FIGURE 19.24a Decomposition of the two-port storage element that represents a motor

The signs of the indicated bond variables show that this bond graph can be simplified into the bond graph in figure 19.24b.

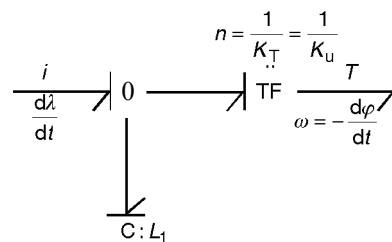


FIGURE 19.24b Simplification of the bond graph of figure 19.24a

The magnetic port can be given preferred integral causality smoothly. The magnetic port can be dualized into an electric port or rather: the coupling between magnetic and electric domain can be represented by an SGY (figure 19.24c).

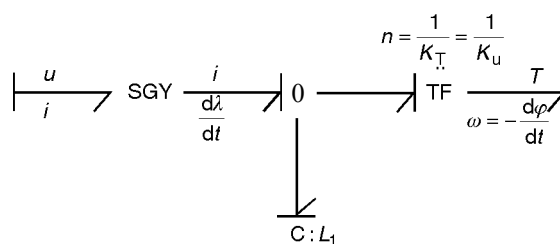


FIGURE 19.24c Coupling between the magnetic and the electric domain by means of an SGY

Next, this SGY can be eliminated by partial dualization. This results in the bond graph model shown in figure 19.24d.

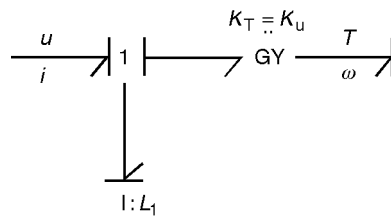


FIGURE 19.24d
Elimination of the SGY by partial
dualization

When it is assumed that storage can be neglected the model of a motor becomes a gyrator. Indeed, chapter 3 already introduced a gyrator for the description of the dominant behavior of an electric motor. However, this assumption is not made here, but the model is extended with rotation–kinetic energy stored in the rotor of the motor. This is achieved by representing the inertia J_m of the rotor by an I–type storage element connected to a 1–junction representing the angular velocity ω of the motor. To this 1–junction an R–element can be connected too to represent the (viscous) bearing friction R_m . Also at the electric port an R–element representing the electric resistance R_e of the windings can be connected to the 1–junction representing the motor current i (figure 19.24e).

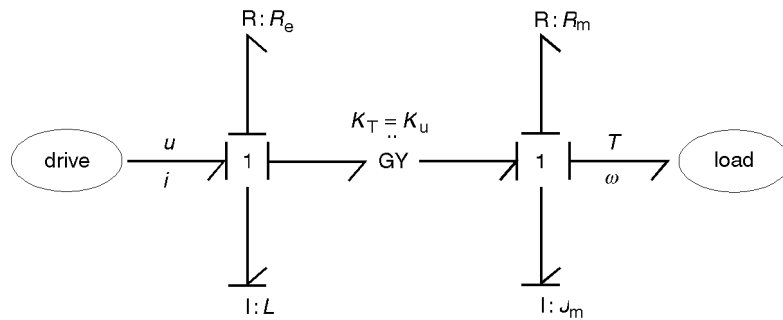


FIGURE 19.24e Bond graph of a electric motor

This is a commonly used motor model, but it is mostly not obtained in the above way. It is commonly based on the Lorentz force, viz. the gyrator model in chapter 3, to which the storing and resistive properties of the electric circuit and the mechanical rotor are added. However, in this process the earlier mentioned constraint on the time scale of the variables is not made explicitly. This can lead to use of the wrong model in principle.

Due to the reversible nature of the transduction in a multiport storage element the mechanical load and electric power supply are fully exchangeable in the obtained model with an electric load and a mechanical power supply respectively. As a consequence, this model can also serve as the model of an electric generator or dynamo —obviously constrained by the same conditions. Using the knowledge about linear analysis in chapter 14 the bond graph in figure 19.24e can be identified as a damped second order system due to the second order causal path between both I–type storage elements via the gyrator and the two first order causal paths between both I–type storage elements at the one hand and both R–elements at the other hand. The transfer between u and ω can be derived directly from the bond graph, for example using transmission matrices.

$$\begin{aligned}
\begin{pmatrix} u \\ i \end{pmatrix} &= \begin{pmatrix} 1 & R_e + sL \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & K_T \\ \frac{1}{K_T} & 0 \end{pmatrix} \begin{pmatrix} 1 & R_m + sJ_m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} T \\ \omega \end{pmatrix} = \\
&= \begin{pmatrix} \frac{R_e + sL}{K_T} & K_T \\ \frac{1}{K_T} & 0 \end{pmatrix} \begin{pmatrix} 1 & R_m + sJ_m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} T \\ \omega \end{pmatrix} \\
&= \begin{pmatrix} \frac{R_e + sL}{K_T} & \frac{(R_e + sL)(R_m + sJ_m)}{K_T} + K_T \\ \frac{1}{K_T} & \frac{R_m + sJ_m}{K_T} \end{pmatrix} \begin{pmatrix} T \\ \omega \end{pmatrix} \quad (19.66)
\end{aligned}$$

This also means that (if R_m also contains R_{load} , such that $T = 0$):

$$\begin{aligned}
H(s) &= \frac{\omega(s)}{u(s)} = \frac{K_T}{K_T^2 + (R_e + sL)(R_m + sJ_m)} = \\
&= \frac{K_T}{K_T^2 + R_e R_m + s(R_e J_m + R_m L) + s^2 L J_m} \approx \frac{1}{K_T} \frac{1}{1 + s R_e J_m K_T^{-2}} \quad (19.67)
\end{aligned}$$

The latter approximation holds for a motor with the (realistic) parameter values in table 19.1:

TABLE 19.1 Realistic parameter values of an electric motor

<i>Motor constants (at 25 °C)</i>	<i>Symbol</i>	<i>Units</i>	<i>Value</i>
Torque constant	K_T	N m A ⁻¹	3.55·10 ⁻²
Back-EMF-constant	K_u	V s rad ⁻¹	3.55·10 ⁻²
Electric resistance	$R_e = R_A + R_T$	Ohm	2.52
Average friction torque	T_F	N m	2.1·10 ⁻²
Viscous damping	$R_m = K_D$	N m s rad ⁻¹	3.97·10 ⁻⁵
Inertia	J_m	N m s ²	1.97·10 ⁻⁵
Self induction of the windings	L	H	< 10 ⁻⁴
Temperature coefficient of K_u	C	% / °C increase	-0.02

The equality $K_T = K_u$ only holds if SI-units are used. If an American handbook is consulted, the torque constant, for example, would be specified in Ounce Inch A⁻¹ and the back-EMF-constant in V per round per minute. This is emphasized, because in practice SI-units are often not used. Also in

Europe motor specifications rounds per minute or RPM are more frequently encountered than rad s^{-1} .

Substitution of the obtained parameter values in the derived transfer results in:

$$H(s) = \frac{3.55 \cdot 10^{-2}}{1.26 \cdot 10^{-3} + 1 \cdot 10^{-4} + s(4.96 \cdot 10^{-5} + 3.97 \cdot 10^{-9}) + 1.97 \cdot 10^{-9} s^2} \approx \quad (19.68)$$

$$\approx \frac{1}{3.55 \cdot 10^{-2}} \frac{1}{3.94 \cdot 10^{-2} s + 1}$$

Note that the accuracy of the data in the specifications is maintained independent of this being realistic.

The approximation of the transfer made earlier for this motor appears to be justified indeed, such that electric resistance, inertia and motor constant determine the dynamic properties (time constant) of this transfer. The *dominant time constant* equals $R_e J_m K_T^{-2} = 0.0394 \text{ s}$. Note that the *mechanical time constant* equals $\tau_m = J_m / R_m = 0.496 \text{ s}$ and the *electric time constant*

$\tau_e = L / R_e = 3.97 \cdot 10^{-5} \text{ s}$, such that a superficial analysis could lead to the wrong conclusion that the mechanical time constant dominates the dynamic behavior of the transfer. Note that for signal frequencies above 2 Hz ($\approx 12.6 \text{ rad s}^{-1}$) the motor cannot be correctly represented by a gyrator only (the effective motor constant K_u' , the inverse gain of the transfer, increases for that frequency already with about 12%). Note that the mechanical resistance not only consists of the rather low viscous friction, but also of a dry friction. When the motor runs, this has the nature of a negative effort source (drain), such that it has no dynamic influence. Both resistance contributions can become considerably larger due to the load.

In practice the motor torque that for one winding gets the form sketched in the solid line in figure 19.25 is given a more constant value by adding more windings (cf. figure 19.25).

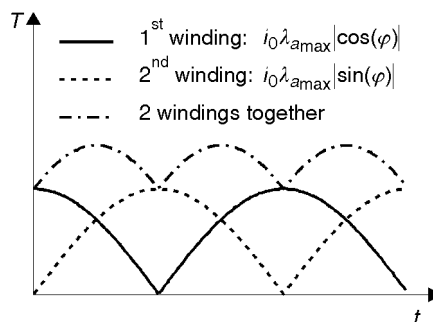


FIGURE 19.25
Motor torque in the case of two windings

Summary

All examples in this chapter demonstrated that two-port and multiport storage element models and cycle processes performed with these models offer an alternative and informative view on the topic of power transduction that is important in engineering. Chapter 21 will treat this issue from yet another angle.

References

Fluitman, J. (1980) [Energy considerations concerning current loops and magnetic objects](#). *American J. of Physics*, 48 (7). pp. 558-562. ISSN 0002-9505

EXPERIMENTS

Perform the experiments in chapter 24 that correspond to chapter 19. If chapter 24 is not (yet) available to you, go to www.20sim.com, download the demo version of 20-sim, study the tutorial and try to use 20-sim for the simple problems of the current chapter where applicable.

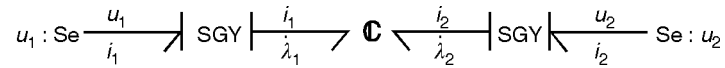
SELF TEST

- 1 Use the characteristics in the solution of assignment 18.8 to design a cycle process with a water container with a massless float that transduces hydraulic power into mechanical power. Impose the changes in V and x in such a way that preferred causality is obtained. Show why storing transducer cannot be a *linear* motor.
- 2 Design a cycle process with the elastic coil that transduces mechanical power into magnetic (electric) power. The constitutive relations of the elastic coil were already derived in the previous chapter (equations (18.21) and (18.22)), but it is preferred to drive the elastic coil by means of a current source, such that $F(i, x)$ is taken as a starting point as obtained in the solution of assignment 18.12.
- 3 Given the bond graph of the model of a DC electric motor in figure 19.24e. Which parameters dominate the transfer during current control?

FEEDBACK

1 Solution of the assignments

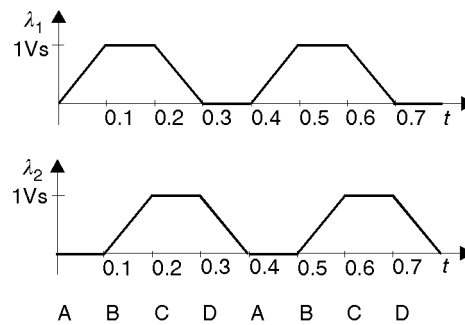
19.1 The causal bond graph is shown in the next figure.



Dualizing the two-port C into a two-port I can also eliminate the SGY's. Both ports of the storage element have integral causality.



In order to find λ_1 and λ_2 , u_1 and u_2 should be integrated with respect to time. Without writing equations, the result can be immediately sketched:



19.2 Assignment (18.15) already showed that the given parameter values do not correspond with integral causality, which requires inversion of the induction matrix. The induction matrix is:

$$\begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

After inversion

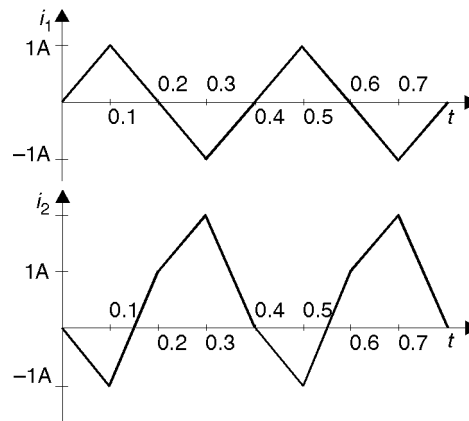
$$\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

such that

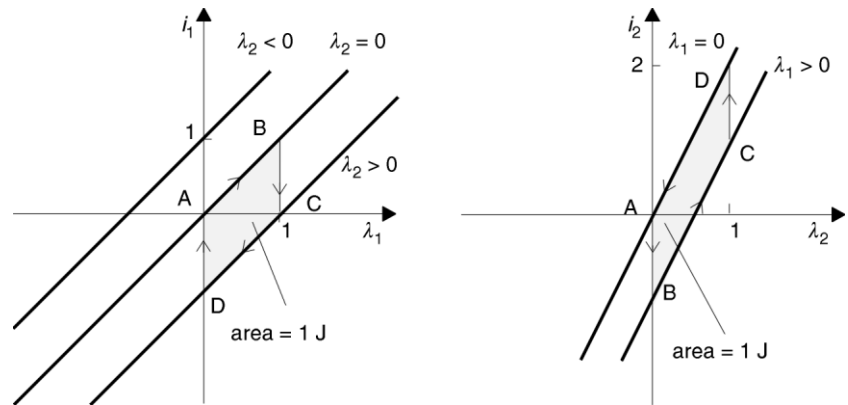
$$i_1 = \lambda_1 - \lambda_2$$

$$i_2 = -\lambda_1 + 2\lambda_2$$

Using the solution of the previous assignment it is found for the four segments of the period:



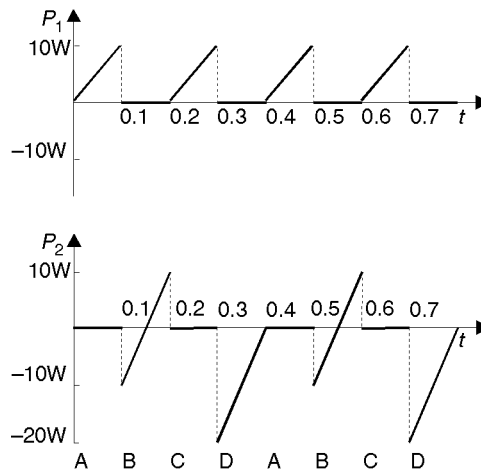
- 19.3 The characteristics have a slope +1 in the i_1 - λ_1 plot (cf. matrix), such that all characteristics are in parallel. For $\lambda_2 = 0$ Vs the characteristic intersects the origin, for $\lambda_2 > 0$ Vs the characteristic intersects the i -axis below the origin and for $\lambda_2 < 0$ Vs above it. The characteristics that are of interest to the cycle are only those for which $\lambda_2 = 0$ Vs and $\lambda_2 = +1$ Vs. The characteristics have a slope +2 in the i_2 - λ_2 plot and furthermore the same can be said as for the first plot.



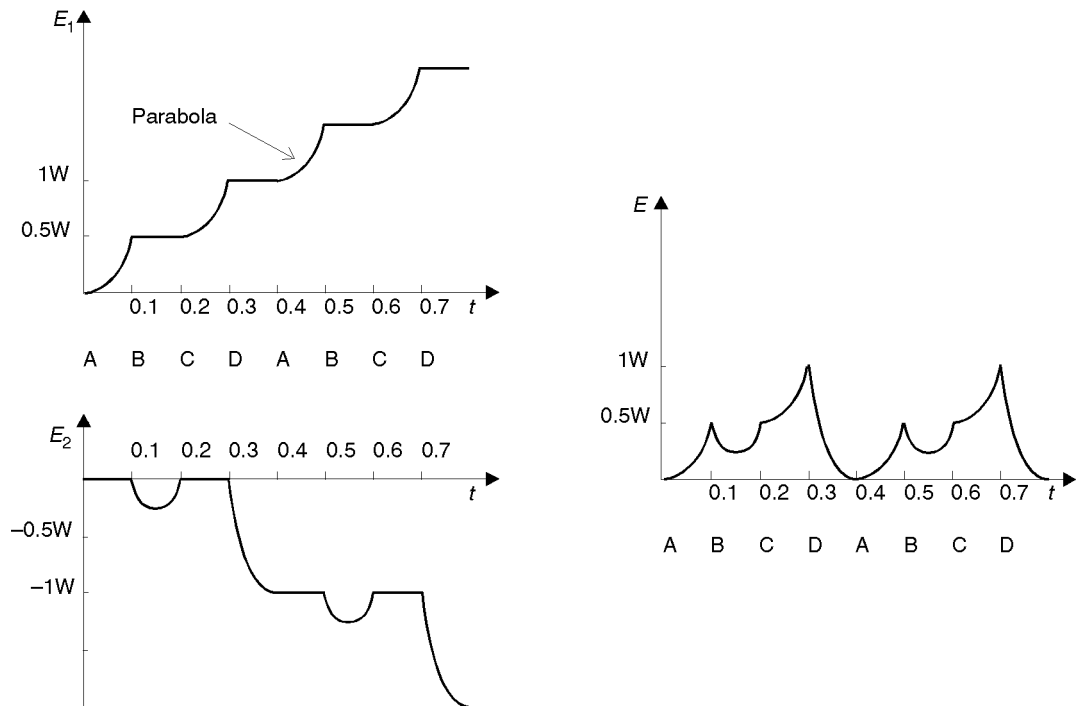
In these plots the cycle ABCD is indicated too. In the first plot it runs clockwise and counterclockwise in the second. The first port absorbs energy during 1 cycle, while the other delivers an amount of energy, both equal to the encircled area, being 1 J.

- 19.4 The input powers as a function of time for the both ports are found from $P_1 = u_1 i_1$ and $P_2 = u_2 i_2$.

The plot of the power in the four sections is shown in the next figure:



The input energy as a function of time for both ports is found from $E_1 = \int P_1 dt$ and $E_2 = \int P_2 dt$. The total amount of energy is $E = E_1 + E_2$. The plot of E_1 , E_2 and E is shown in the next figure.



- 19.6 The constitutive relations of all elements in the decomposition are written (the Laplace operator is used for the linear storage element relations) and next all internal variables are eliminated, such that a direct description of the port behavior remains:

$$e_a = e_{C_a} - e_{TF_2}$$

$$f_{C_a} = f_a$$

$$f_{TF2} = f_a$$

$$se_{C_a} = \frac{f_{C_a}}{C_a}$$

$$e_{TF2} = \frac{1}{n} e_{TF1}$$

$$f_{TF1} = \frac{1}{n} f_{TF2}$$

$$se_{C_b} = \frac{f_{C_b}}{C_b}$$

$$f_{C_b} = f_b - f_{TF1}$$

$$e_b = e_{C_b}$$

$$e_{TF1} = e_{C_b}$$

or, after elimination of the internal variables:

$$se_a = \frac{f_a}{C_a} - \frac{1}{n} se_b = \frac{f_a}{C_a} - \frac{1}{n} \left(\frac{-f_a}{nC_b} + \frac{f_b}{C_b} \right) = \left(\frac{1}{C_a} + \frac{1}{n^2 C_b} \right) f_a - \frac{1}{nC_b} f_b$$

$$se_b = \frac{f_b - \frac{1}{n} f_a}{C_b} = -\frac{1}{nC_b} f_a + \frac{1}{C_b} f_b$$

This can be written in matrix form:

$$s \begin{pmatrix} e_a \\ e_b \end{pmatrix} = \begin{pmatrix} \frac{1}{C_a} + \frac{1}{n^2 C_b} & \frac{-1}{nC_b} \\ \frac{-1}{nC_b} & \frac{1}{C_b} \end{pmatrix} \begin{pmatrix} f_a \\ f_b \end{pmatrix}$$

This is result identical to the constitutive relations of a two-port storage element with constitutive matrix C :

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{pmatrix}$$

for which holds

$$s \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \frac{C_{22}}{C_{11}C_{22} - C_{12}^2} & \frac{-C_{12}}{C_{11}C_{22} - C_{12}^2} \\ \frac{-C_{12}}{C_{11}C_{22} - C_{12}^2} & \frac{C_{11}}{C_{11}C_{22} - C_{12}^2} \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

This means that both two-ports are equivalent if

$$\left(\frac{1}{C_a} + \frac{1}{n^2 C_b} \right) = \frac{C_{22}}{C_{11}C_{22} - C_{12}^2}$$

$$-\frac{1}{nC_b} = \frac{-C_{12}}{C_{11}C_{22} - C_{12}^2}$$

$$\frac{1}{C_b} = \frac{C_{11}}{C_{11}C_{22} - C_{12}^2}$$

or:

$$C_b = C_{22} - \frac{C_{12}^2}{C_{11}} = C_{22} \left(1 - \frac{C_{12}^2}{C_{11}C_{22}} \right) = C_{22} (1 - k^2)$$

$$n = \frac{C_{11}}{C_{12}} = \frac{\sqrt{C_{11}C_{22}}}{C_{12}} \sqrt{\frac{C_{11}}{C_{22}}} = \frac{1}{k} \sqrt{\frac{C_{11}}{C_{22}}}$$

$$C_a = C_{11}$$

This is indeed shown in figure 19.2a,b, such that this decomposition is equivalent as for port behavior with the linear two-port storage element.

19.7 By definition: $\lambda_a = \lambda_1|_{i_1=0} = 0L_1 + Mi_2 = Mi_2$

19.8 Suppose $\lambda_a = A/(x^2 + B)$, where A is the (positive) proportionality constant and B a positive constant to prevent that the applied flux would become unrealistically large in the neighborhood of $x = 0$, the position where the magnet is in the middle of the coil, then

$$F = \frac{-\partial \tilde{E}^*}{\partial x} = \frac{-\partial}{\partial x} \left(\frac{L_1 i_1^2}{2} + i_1 \lambda_a(x) \right) = \frac{-\partial}{\partial x} \left(\frac{L_1 i_1^2}{2} + i_1 \frac{A}{x^2 + B} \right) = i_1 \frac{2xA}{(x^2 + B)^2}$$

The partial derivative of F with respect to x becomes

$$\frac{\partial F}{\partial x} = i_1 \left(\frac{2A}{(x^2 + B)^2} - 4x \frac{2xA}{(x^2 + B)^3} \right) = 2Ai_1 \frac{x^2 + B - 4x^2}{(x^2 + B)^3} = 2Ai_1 \frac{B - 3x^2}{(x^2 + B)^3}$$

The sign of the partial derivative of the force equals the sign of the current i_1 in the neighborhood of $x = 0$ where $B > 3x^2$. This corresponds to the experience that (electro-)magnets attract each other in case of equal flux directions and repel each other in case of opposite flux directions.

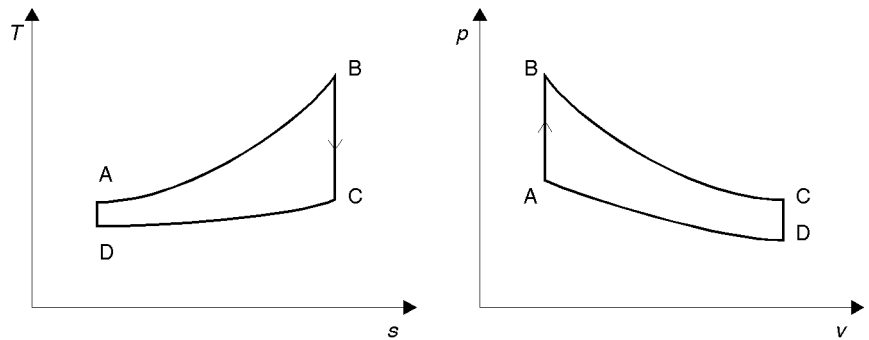
This means that the system for $i_1 > 0$ (attracting state) is intrinsically stable, i.e. it behaves like a spring, but in case $i_1 < 0$ (repelling state) it is intrinsically unstable: the system parts 'fly apart'. Note that motions in another direction than the x -direction are assumed to be zero, for instance in case of a guiding rail.

- 19.9 For a specific value of $i \neq 0$ this two-port is intrinsically stable when $dF/dx > 0$:

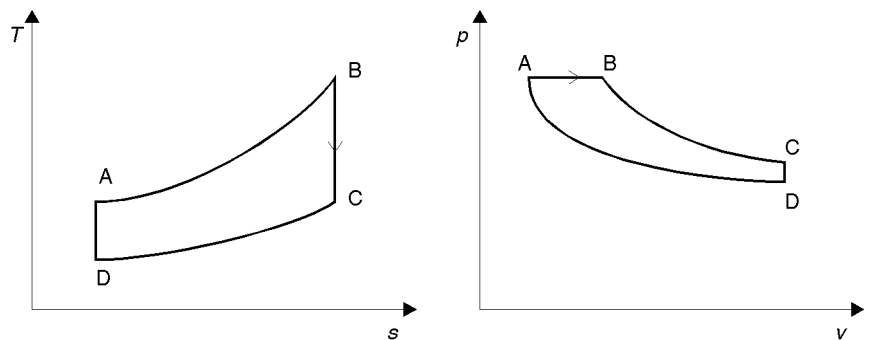
$$\frac{dF}{dx} = \frac{dF}{dX} \frac{dX}{dx} = (L_{\max} - L_{\min}) \frac{2i^2}{x_{\max}^2} (1 - 3X^2)$$

which demonstrates that this system only behaves like an intrinsically stable spring when $|X| < (1/3)^{0.5}$ or $|x| < x_{\max}(1/3)^{0.5}$. In other words: inside the stability area the core will return into the coil after being released, but beyond the stability boundary it will be launched from it.

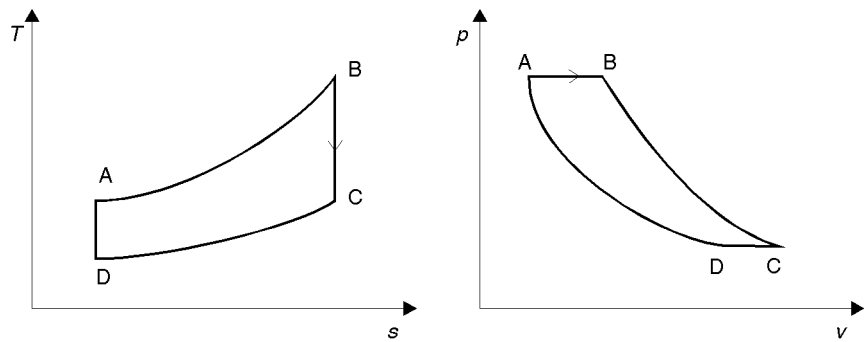
- 19.10 The different cycles are shown graphically in the following figures.



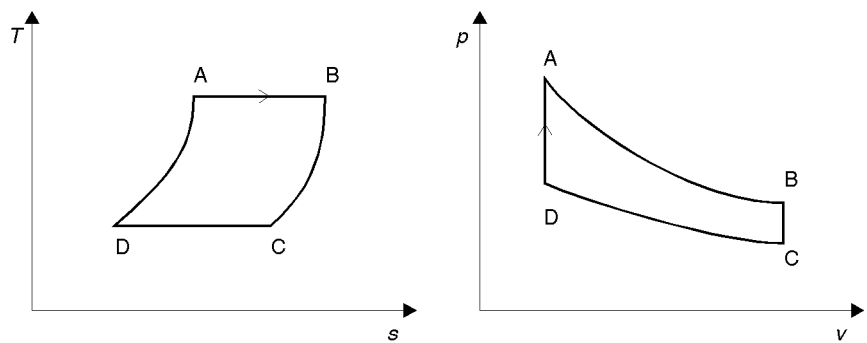
The Otto cycle



The Diesel cycle



The Joule–Brayton cycle



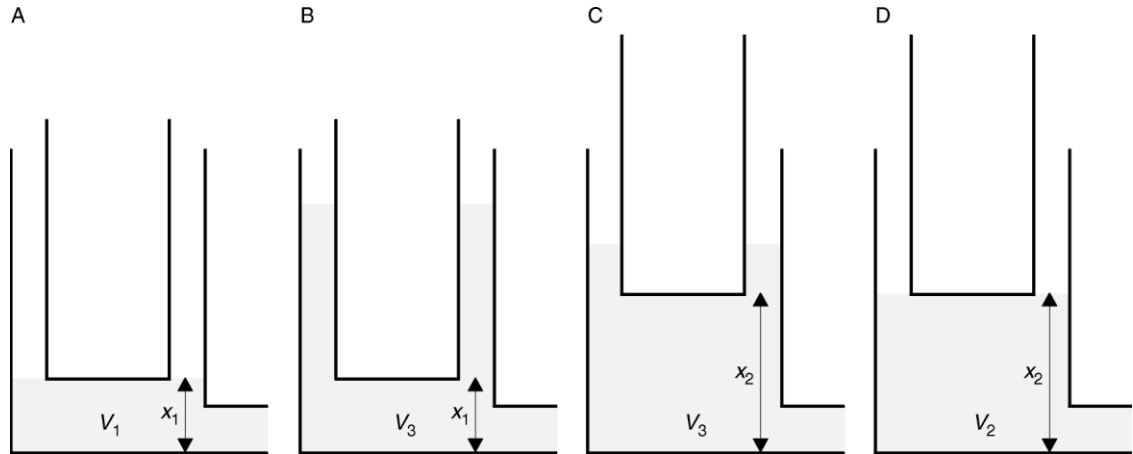
The Stirling cycle

- 19.11 A short summary of the operation of a cooling system in a domestic refrigerator is that the process is based on a vapor–compression system. The working fluid (medium), which used to be Freon, but has now been replaced by environmentally safer alternatives is compressed by a motor with compressor that are both contained in a sealed tank to prevent leakage. In the condenser the compressed working fluid transfers heat to the environment and becomes a liquid, next it is expanded (via an expansion valve) while extracting (evaporation) heat via the cooling element from the interior of the refrigerator. Freon has a number of properties that make it extremely suitable as a working fluid in this kind of cooling cycle: a low surface tension, a low boiling temperature (important for extracting evaporation heat at a temperature of a few degrees Celsius) and a low viscosity (important for a low flow resistance in the system). Furthermore, it is a quite stable species that is not (directly) dangerous for health, not inflammable, not aggressive for the materials that it is usually in contact with and it does not result in a detectable smell in the cooling system. Apart from the suspected relation to the degradation of the ozone layer it is the almost perfect working fluid in a cooling cycle.

2 Solution of the self test

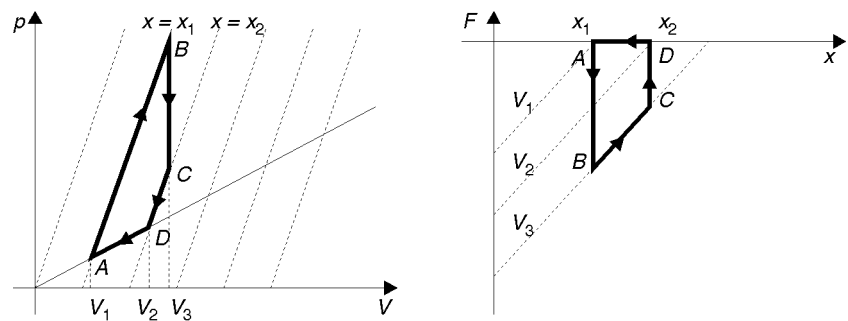
- 1 The constitutive relations of a water container with massless float are derived already in the previous chapter (equations 18.13a and b). The bundles of characteristics were already sketched too (solution to assignment 18.8). It can be easily seen that if the bottom of the float reaches the bottom of the container or rises higher than the rim of the container the two–port ceases to be a two–port, such that a linear transducer cannot be realized. A possible cycle process that

tries to achieve a large enclosed area is the process for which the different stages are sketched in the following figure. (The two-port storage element maintains its preferred causality during this cycle process. This means that the changes of V and x are imposed.)



- A $V = V_1, x = x_1$ (where $V_1 = Ax_1$: float rests at water surface)
 $A \rightarrow B$ container filled to V_3 , with position of float fixed in x_1
 B $V = V_3 > V_1, x = x_1$
 $B \rightarrow C$ float allowed to move upward to x_2 and fixed there
 C $V = V_3, x = x_2 > x_1$
 $C \rightarrow D$ container empties until float just touches the surface
 D $V = V_2 = Ax_2, x = x_2$
 $D \rightarrow A$ container empties further to V_1 , while the float remains touching the surface and consequently also moves to x_1

The cycle turns clockwise in the p - V plane and counterclockwise in the F - x plane such that hydraulic power is transduced into mechanical power.

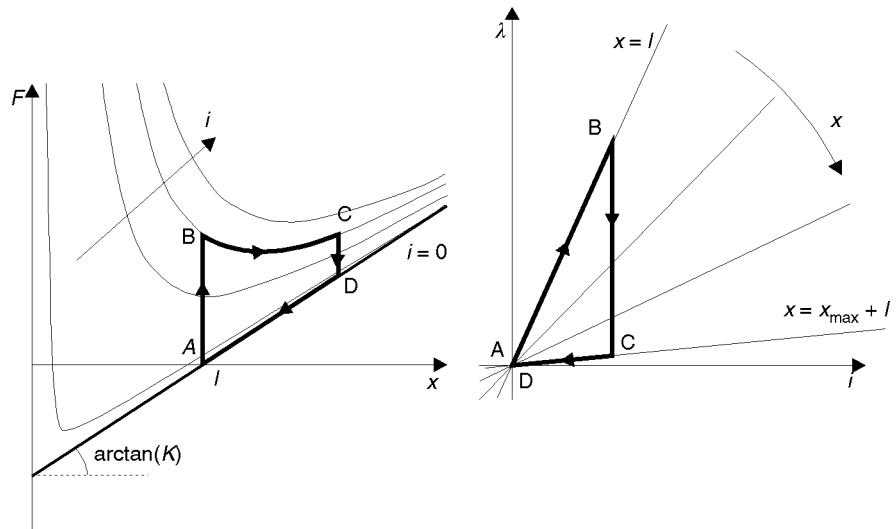


2 A possible cycle process that tries to achieve a large enclosed area is the following:

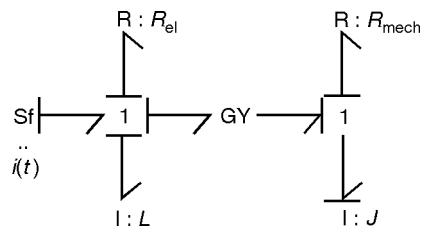
- A $x = l$, the rest length of the coil, $i = 0$
 $A \rightarrow B$ length coil is fixed, current source is switched on
 B $x = l, i = i_{\max}$
 $B \rightarrow C$ the spring is extended until $l + x_{\max}$, while the current source remains active
 C $x = l + x_{\max}, i = i_{\max}$

- $C \rightarrow D$ the length of the coil is fixed and the current source is switched off
 D $x = l + x_{\max}$, $i = 0$
 $D \rightarrow A$ the spring is released again, while the current source remains deactivated

The cycle turns clockwise both in the F - x plane and in the λ - i plane. In order to achieve integral causality, the i - λ plane should be drawn. In that case the cycle would turn counterclockwise, such that mechanical power is transduced into magnetic and consequently electric power.



- 3 The bond graph of an electric motor with current control is shown in the next figure.



Due to the current control the electric I-element obtains differential causality. The current and (via the gyrator) the motor torque are imposed. The behavior of the motor thus becomes completely determined by parameters of the gyrator and of the mechanical part of the motor, i.e.: by the motor constant and by the inertia J and the friction R_{mech} .